



SECJ2203: Software Engineering

System Documentation (SD)

KADA Corporation System

Version 1.0

12/12/2024

Faculty of Computing

Prepared by: CO-T

Revision Page

a. Overview

In this System Documentation (SD), a detailed documentation of the automated cooperative system for Koperasi KADA will be provided. The documentation consists of the introduction that comprises the purpose and scope of the system. The documentation will also provide the specific requirements that consist of user persona, user story and system feature. The documentation will be supported with different use case diagrams. The documentation also covers the non-functional and the design constraints imposed by Koperasi KADA.

b. Target Audience

The target audience for this system is the non-member, member, staff and Ahli Lembaga Koperasi (ALK) of Koperasi KADA.

c. Project Team Members

Table xxxx below shows the task distribution for each member in completion of SD. The table shows the roles and status for each assigned task for each member by sections for this SD version (complete, partially complete, incomplete).

Member Name	Role	Task	Status
Nurul Ika Syafiny Binti Azhar	Team Leader	Project planning	Complete
		Project management	Incomplete
		Develop system	Incomplete
Nuraisyah Binti Mohd Zikre	Technical Writer	Gather user requirements	Complete
		Create System Requirement Specification	Complete
		Create system documentation	Incomplete
Harini A/P Sangaran	System Analyst	Create system architecture	Complete
		Design system functional diagram	Complete
		System Implementation	Complete
Damiya Aina Binti Basir Abd Shammad	System Designer	System interface design	Complete
		System testing	Incomplete

d. **Version Control History**

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Nurul Ika Syafiny Binti Azhar	Completed Chapter 1, Section 1.1 Chapter 2, Section 2.1.4, 2.2.1, 2.4.4, 2.6	20/12/2024
1.0	Nuraisyah Binti Mohd Zikre	Completed Revision Page, Chapter 1 (Introduction), Section 1.3, Section 2.1.1, Section 2.2.2, Section 2.4.1, Section 2.6	21/12/2024
1.0	Harini A/P Sangaran	Completed section 1.2, section 2.1.2, section 2.2.2, section 2.4.2, and section 2.5	21/12/2024
1.0	Damiya Aina Binti Basir Abd Shammad	Complete section 2.1.3, section 2.3 and also 2.5	22/1/2024

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Introduction

In this System Documentation (SD), the detailed information of the system that has been prepared by the team member after the discussion will be included. In this documentation, it consists of the details of Software Requirements Specification (SRS), Software Design Description (SDD) and Software Test Documentation (STD). Each documentation of SRS, SDD and STD will provide a better understanding in development of the system and the implementation of the system in order to achieve a system that satisfies the target user.

1.1 Purpose

This System Documentation (SD) describes the system's users, functions and features in detail to improve better understanding of the system. This is to help the developer grasp clearly the structure of the system and system flow. This SD includes system details such as Use Case Diagram, Class Diagram and State Diagram further help designer and developer get the clear picture of the system through diagram. Moreover, this SD will also include the functional and non-functional requirements so that the designers can design a system that fulfills them all and the tester can check if all the requirements are satisfied. This is also for client reference so that they can add up the missing functions and give feedback about the planned system before moving further to the implementation phase. Lastly, this SD will also include the design constraints so that the team can focus on it and work together to find a solution.

1.2 Scope

The software product that is used in the proposed system is phpMyAdmin MySQL. MySQL is used to store the databases that are used in the proposed system. For example, the proposed system for Koperasi KADA has multiple databases to store information for non-members, members, staff and Ahli Lembaga Koperasi (ALK). Storing the data in databases is crucial so that the information can be used for verification and also can be stored for the future purposes of staff and ALK. Using databases in the proposed system is efficient because the user details are interrelated. For example, the membership application list will be used both by the staff and ALK. Hence, the retrieval of user information will be easier, time efficient and any modification made to the user information has to be made once only at the database.

Besides that, XAMPP Control Panel is the software used as the web server for the proposed system. XAMPP Has multiple modules under its software, such as Apache, MySQL, FileZilla, Mercury and Tomcat. For our proposed system, only Apache and MySQL will be

used. Apache functions as the web server, enabling the execution of PHP scripts, which are developed using Visual Studio Code.

1.3 Definitions, Acronyms and Abbreviation

This subsection will provide the definitions of all terms, acronyms, and abbreviations that will be used in this documentation.

- ALK: The board members that were responsible for approving the application applied.
- Non-member: A person who is not a member of Koperasi KADA
- Member: A person is a member of Koperasi KADA
- Staff: An employee in Koperasi KADA who will review the application and send it to ALK
- SD: System Documentation

1.4 References

[1] UXPin. (2023, January 23). *7 Constraints in Design and How to Overcome Them*. Studio by UXPin. <https://www.uxpin.com/studio/blog/constraints-in-design/>

1.5 Overview

The entire process of designing and developing the proposed Koperasi KADA system is described in this System Documentation (SD). It outlines the team's approach for resolving the inefficiencies of the current manual procedures by using a web-based, digital platform that is customized to meet the requirements of the cooperative. The goal, requirements, design, and testing techniques of the system are all included in the multiple parts that make up the SD. Each part verifies that the system's usability, functionality, and operational effectiveness are thoroughly documented for easy setup and maintenance.

The System Documentation is organized to provide a clear and logical progression through the system development process. It begins with the System Introduction, which outlines the objectives and rationale for developing the Koperasi KADA system, focusing on addressing the inefficiencies of the current manual processes. The system's functional and non-functional requirements are then described in the Specific Requirements part, assisting in defining the system's goals and parameters. In order to make sure that the system design meets the demands of the target users, this part contains a Persona subsection that characterizes them.

The main features of the system, including online membership registration, loan applications and real-time financial tracking, are then covered in detail in the System Features section.

The Launch Phase then outlines the deployment plan, including how to roll out the system, onboard users, and replace the manual processes that are currently in place. Additionally, use case descriptions that show how users will interact with the system in different scenarios are included in the User Story Details section. These explanations help to make sure that every feature successfully satisfies user wants and is in line with the system's overall objectives. In order to guarantee that the platform operates at its best in a variety of circumstances, the Performance and Other Requirements section also concentrates on technical and operational factors like system scalability, dependability, and response time.

Finally, the Design Constraints section highlights the limitations and specific boundaries within which the system must operate, ensuring the design remains feasible and realistic. By organizing the document into these sections, the SD provides a clear and structured roadmap for the system's development, ensuring that all stakeholders can contribute effectively to achieving a successful implementation.

Specific Requirements

This section of SD contains all the details of software requirements such as Persona and Use Case Diagram of the system to enable designers to design the system that satisfies all the requirements. Throughout this section, every stated requirement is explained clearly in many kinds of forms and diagrams to enhance the understanding of users, operators and other external systems. These requirements also include a description of every input into the system, every output from the system, and all functions performed by the system in response to an input or in support of an output.

2.1 Persona

In this subtopic, we will describe the intended users of the system and their characteristics to provide the developer with a clear understanding of whom the system is designed for, and helps to ensure that the system is developed to meet the needs of its intended user. This system has four types of users which are non-member, member, staff and Ahli Lembaga Koperasi. The details of these actors will be explained using persona as in below.

2.1.1 Persona 1 (Member/Actor 1)

User Need

- A member needs a way to login to the system to apply for a loan, view loan application approval, loan balance, savings and financial statement.

US001 - User Stories

- As a member, I want to login to the system so that I can apply for a loan, view my loan application result whether it is approved or not, view the balance of loan that needs to be paid, view the savings from the deduction of the membership fee every month and check the payment statement.

A member apply for loan
To apply for a loan you must first open the website of Koperasi KADA and login into the system. If you did not log in into the system, you cannot apply for a loan. After successfully logging into the system, you can apply for a loan after clicking on the “Mohon Pinjaman”. You are required to fill in the required details in the form in order to submit the application. If you do not fill in the required details, you are unable to submit the form.

After successfully submitting the form, the application will be processed. In all cases, you need to log in into the system in order to apply for a loan.

Table xxx: A member user story 1

A member view loan application approval

If you apply for a loan, you can check your loan application approval by clicking on the “Semak Kelulusan Pinjaman”. If your application is approved, the details of the application will be displayed.

If you did not apply for a loan, the system will show that you did not apply for a loan.

In all cases, you need to log in into the system in order to view loan application approval.

Table xxx: A member user story 2

A member view loan balance

If you apply for a loan and the application is approved, you can check your loan balance by clicking on the “Baki Pinjaman”. The balance of the loan that needs to be paid will be displayed by the system.

If you did not apply for a loan, the system will show that you did not balance that need to be paid.

In all cases, you need to log in into the system in order to view loan balance.

Table xxx: A member user story 3

A member view savings

If you want to view savings, you can click on the “Semak Simpanan”. The details of the savings will be shown by system.

In all cases, you need to log in into the system in order to view savings.

Table xxx: A member user story 4

A member view financial statement
If you want to view the financial statement, you can click on the “Semak Penyata”. The system will display the details of every transaction of money for every month. In all cases, you need to log in into the system in order to view savings.

Table xxx: A member user story 5

2.1.2 Persona 2 (Staff/Actor 2)

User Need

A user who is a staff of Koperasi KADA needs to be able to view the list of Koperasi KADA members and Koperasi KADA members who have a loan with Koperasi KADA and validate the verification documents of users who applied for membership and loan.

US002 - User Stories

As a staff, I want to validate the verification documents of non-members who applied for a membership and I want to validate the verification documents of members who applied for a loan.

A staff logging into the system
If you want to login to the system, you must first open the website of Koperasi KADA to login to the system as a staff. Then, you have to key in your staff ID and password to login to the system.

Table xxx: A staff user story 1

A staff viewing list of Koperasi KADA members
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership and list of members who applied for loans. If you click to view the list of Koperasi KADA members, a list of members along with their ID number will be displayed. You have an option to download the members list if you want to.

Table xxx: A staff user story 2

A staff viewing list of Koperasi KADA members who applied for loan
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership and list of members who applied for loans. If you click to view the list of Koperasi KADA members who have a loan with Koperasi KADA, a list of members along with their ID number and the type of loan they took will be displayed. You have an option to download the members with loans list if you want to.

Table xxx: A staff user story 3

A staff validating the verification documents of non-members who applied for membership
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership and list of members who applied for loans. If you click to view the list of non-members who applied for membership, a list of users with their ID numbers and verification documents will be displayed. You can click on the verification documents to check the documents. If the documents are as per requirements, you can validate the verification documents. Once you have validated the verification documents, the validation status of the documents will be updated. You can also choose to update the validity of the verification documents later on if there's any changes.

Table xxx: A staff user story 4

A staff validating the verification documents of members who applied for loan
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership and list of members who applied for loans. If you click to view the list of members who applied for a loan, a list of members with their ID numbers and verification documents will be displayed. You can click on the verification documents to check the documents. If the documents are as per requirements, you can validate the verification documents. Once you have validated the verification documents, the validation status of the documents will be

updated. You can also choose to update the validity of the verification documents later on if there's any changes.

Table xxx: A staff user story 5

2.1.3 Persona 3 (ALK / Actor 3)

User Need

A user who is on the high committee in Koperasi KADA needs a way to verify the membership applications by a non-member and also the loan applications by a member of Koperasi KADA.

US003 - User Stories

As a high committee member, I want to verify the membership applications of non-members so that I can ensure only eligible applicants become members of Koperasi KADA.

An ALK log into the system
If you want to login to the system, you must first open the website of Koperasi KADA to login as a high committee member. Then, you have to key in your ID and password to access the system.

Table 001: ALK User Story 1

As a high committee member, I want to verify the loan applications of members so that I can approve loans for eligible members of Koperasi KADA.

An ALK viewing the membership application
Once logged into the system, you will be able to view the homepage with features to view: 1. Membership applications from non-members. 2. Loan applications from members.

Table 002: ALK User Story 2

As a high committee member, I want to review the verification documents of non-members applying for membership to validate their eligibility.

An ALK validating the membership application
If you click on the list of non-members applying for membership, a table will be displayed with their ID numbers and verification documents. You can open the documents to verify their eligibility. Once verified, the status will be updated in the system. You can also update the status later if there are any changes.

Table 003: ALK User Story 3

As a high committee member, I want to review the verification documents of members applying for loans to validate their eligibility.

An ALK validating the Loan Application
If you click on the list of members applying for loans, a table will be displayed with their ID numbers, loan types, and verification documents. You can open the documents to verify their eligibility. Once verified, the status will be updated in the system. You can also update the status later if there are any changes.

Table 004: ALK User Story 4

2.1.4 Persona 4 (Non-member/Actor 4)

User Need

A user who is a non-member of Koperasi KADA needs a way to become a member of Koperasi KADA.

US004 - User Stories

As a non-member user, I want to become a member of Koperasi KADA so that I can log into the KADA Corporation system.

A non-member apply to be a member
You must first open the website of Koperasi KADA to see the application for membership function. Click on the apply membership at the website to be directed to the application forms. Once you see the application forms, you need to complete all details in their forms and upload the mandatory documents to the system. You are required to make a mandatory payment to apply as a member for the first time. Once the payment is made and the receipt is uploaded, you can submit your application to the system. Once you click 'Submit Application', your form will be recorded in the system for further processing by Koperasi KADA staff and ALK to approve your application . In seven working days, you will get an email about their membership approval status either success or fail. If your application is approved, you will get your member number and password to log in into the system. If you fail, you will be notified through email too and asked to resubmit your application if you want.

A non-member who just become a member want to login for the first time
Once you get login details through email, press the 'Daftar Masuk' button at the homescreen to be directed to the login page of the system. Choose 'Ahli' at the button below the 'Log Masuk' button, then insert the login and password details given in the approval email. If login success, you will be directed to the main tab where all the member functions will be in display and available to use. You are now a Koperasi member. If you fail, try login again or contact the staff for further assistance.

2.2 System Features

The KADA Corporation System is a software system that operates on a web site. The system provides a means for Koperasi KADA members to get updates fast, apply to member's loan easier and get the details for transactions made. It also helps the staff to manage corporation work like the application forms easier and faster than the manual method.

2.2.1 Use Case Diagram

The system features are illustrated in the use case diagram at Figure 1.0 below. The detailed description of each module and functions is tabulated in Table 1.0.

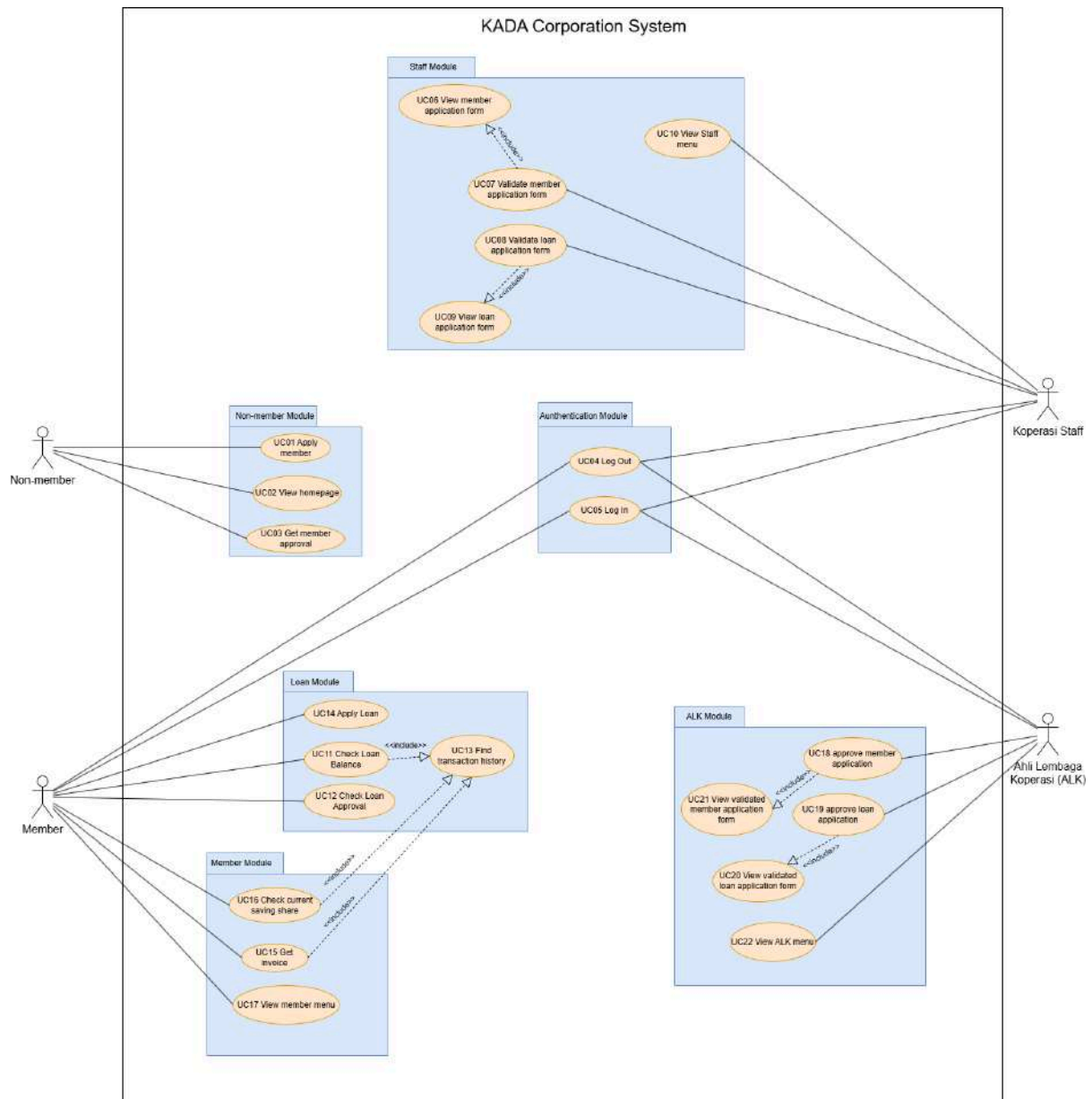


Figure 1: Use Case Diagram for KADA Corporation System

Table 1: Description of Module and Functions for KADA Corporation System

Use case	Function	Description
UC01	Apply member	This use case allows users to apply for Koperasi KADA membership.

UC02	Get member approval	This use case will notify users when their membership application is approved and provide login details to them
UC03	View Homepage	This use case allows all users to see the homepage of the KADA website without logging in.
UC04	Log out	This use case allows users to exit the system.
UC05	Log in	This use case allows users to access the system.
UC06	View member application form	This use case lets users see the submitted membership application form.
UC07	Validate member application form	This use case is for users to validate each submitted membership application form and make sure the details are true before being sent to the ALK.
UC08	Validate loan application form	This use case is for users to validate each submitted loan application form and make sure the details are true before being sent to the ALK.
UC09	View loan application form	This use case lets users see the submitted loan application form.
UC10	View staff menu	This use case allows staff to see the menu of the system where all their functions are available there.
UC11	Check loan balance	This use case lets users check their loan balance.
UC12	Check loan approval	This use case notify users about the status of their loan applications approval.
UC13	Find transaction history	This use case sends all the transaction history to users.
UC14	Apply loan	This use case allows users to send their loan application form for approval.
UC15	Get invoice	This use case prints out the invoice of every transaction made to users.
UC16	Check current saving share	This use case allows users to see their current share and savings.
UC17	View member menu	This use case allows members to see the menu of the system where all their functions are available there.

UC18	Approve member application	This use case allow users to approve the membership application and trigger application status
UC19	Approve loan application	This use case allow users to approve the loan application and trigger application status
UC20	View validated loan application form	This use case lets users see the submitted and validated loan application form.
UC21	View validated member application form	This use case lets users see the submitted and validated membership application form.
UC22	View ALK menu	This use case allows ALK to see the menu of the system where all their functions are available there.

The system features are illustrated in the use case diagram at Figure 1. The detailed description of each module and functions is tabulated in Table 1.

2.2.2 Domain Model: Class Diagram

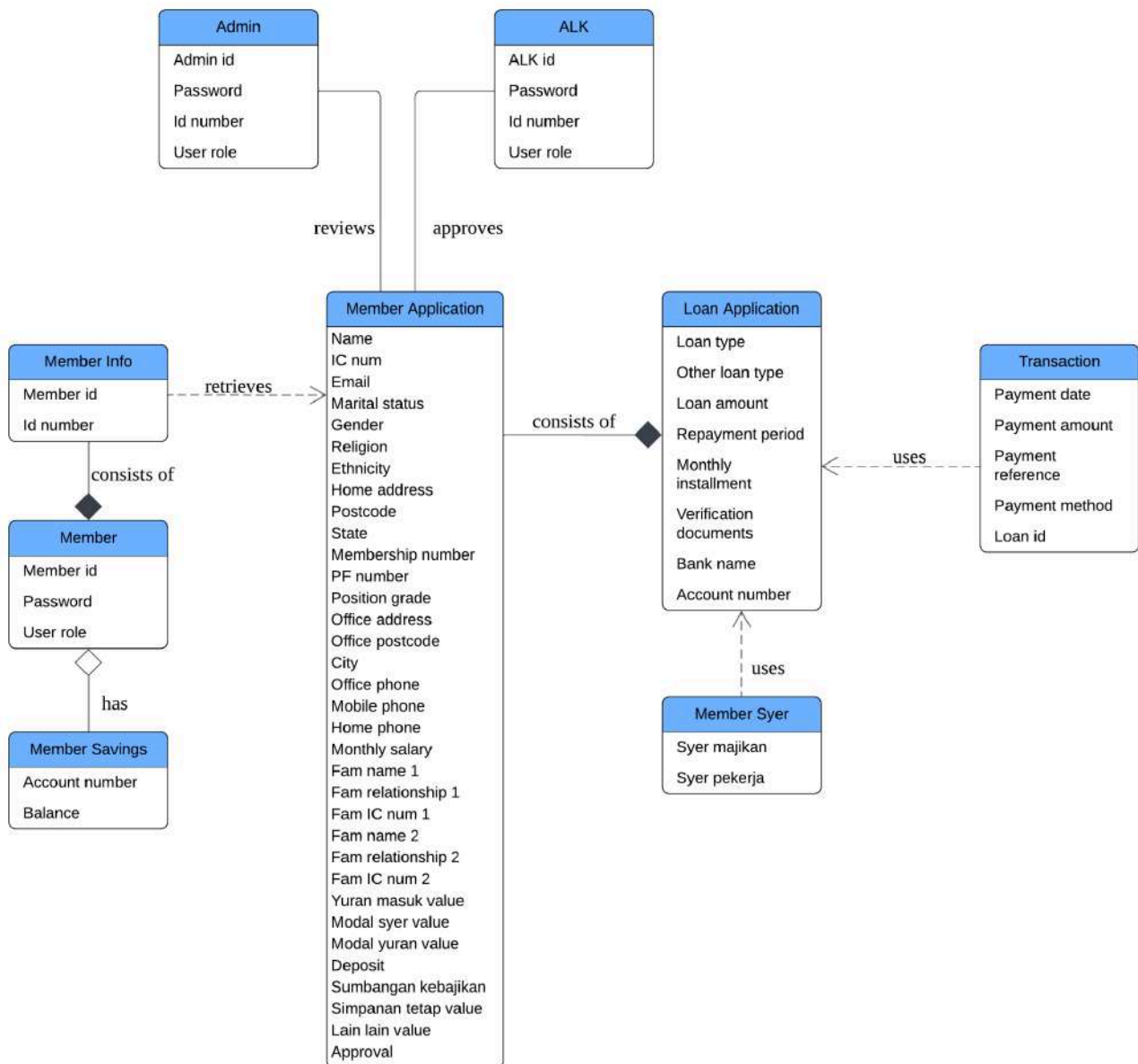


Figure 2: Domain Model for Koperasi KADA system

The classes used in the Koperasi KADA system are illustrated in the domain model at Figure 2 along with the relationship for each class. Each member consists of Member Info table which has member id and id number. Member Info table retrieves member information from Member Application table using id number. A member can have savings which the information will be stored in Member Savings table. A member can apply for loans in which the information will be stored in Loan Application table. Loan Application table consists of member information retrieved from Member Application table. If a member has any loans, then Member Syer and Transaction depends on the loan. Admin and Ahli Lembaga Koperasi (ALK) review and approve the member application respectively.

2.2.3 State Diagram

Figure xxx below shows the main flow of the system where a non-member of Koperasi KADA wants to apply for a membership. The flow starts by a non-member to fill in the form provided and after the submission of the form, it will be reviewed by the staff of Koperasi KADA. If the submitted form has been reviewed and approved by staff, the process will continue by the ALK to review the form. If ALK approves the application, non-member will change their status to member of Koperasi KADA upon the membership is approved by ALK. For a disapproved application, non-member can apply for a membership again by filling in the form again and going through the same process again. Same cycle happens for Figure xxx for a member of Koperasi KADA to apply for a loan where the difference is the loan application can only be applied for member of Koperasi KADA only.

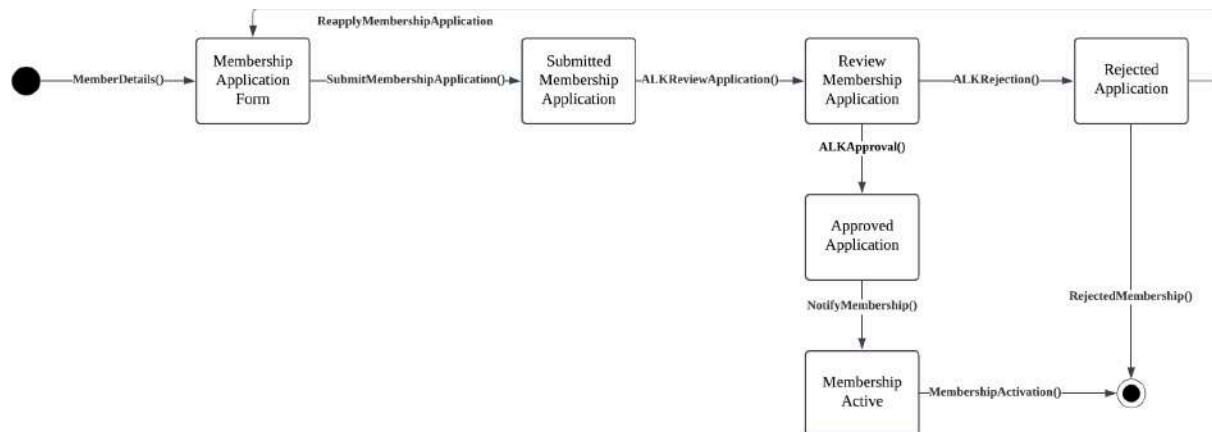


Figure 3: State Machine Diagram for <Name of the System> - syah

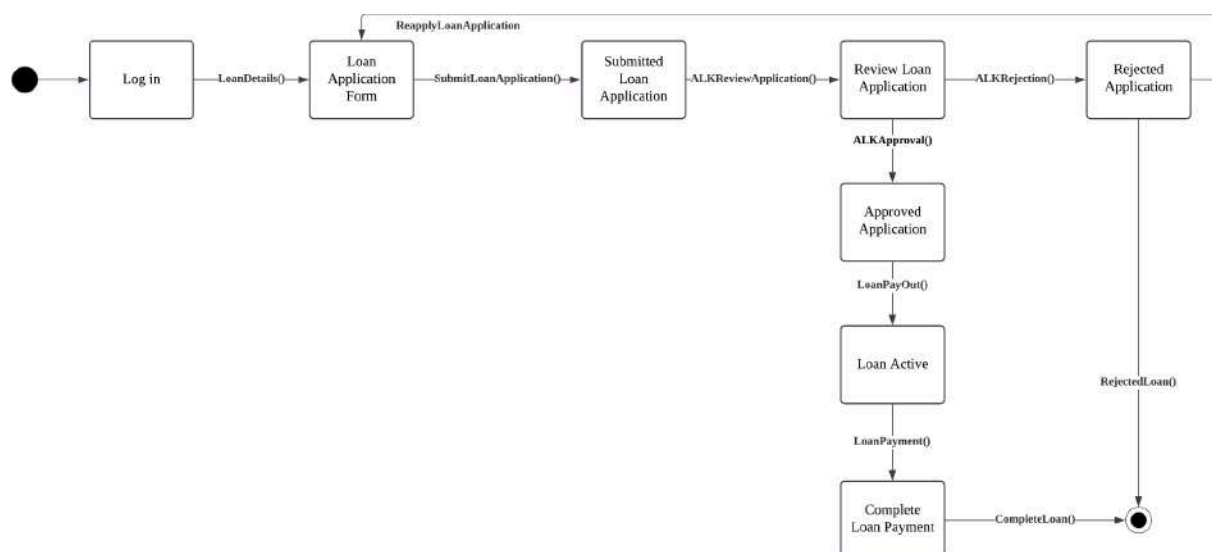


Figure 4: State Machine Diagram for <Name of the System> - syah

2.3 Launch Phase

Backlog Items:

1. Develop a non-member module to allow non-members to apply for membership and track their application status.
2. Develop a member registration and login module to enable new users to register and access the system securely.
3. Implement a loan application and approval module to allow members to submit loan applications and check approval statuses.
4. Create a staff dashboard that facilitates membership and loan validation processes efficiently.
5. Build an ALK dashboard to enable high committee members to review and approve membership and loan applications.
6. Incorporate security and encryption measures to protect sensitive data and ensure compliance with privacy standards.
7. Optimize system performance to handle simultaneous user requests and maintain quick response times.
8. Provide user training and documentation to ensure smooth adoption and usability of the system.

Sprint	Team members assigned
Sprint #1: Non-Member Module Development ● User Stories Covered: ○ US004: Non-member applies for membership. ● Tasks: ○ Develop membership application forms for non-members. ○ Implement functionality to upload required verification documents. ○ Enable non-members to track application status and receive notifications.	Nurul Ika Syafiny binti Azhar

○ Test non-member functionalities to ensure smooth operations.	
Sprint #2: Member Module Development ● User Stories Covered: ○ US001: Member login and registration. ○ US001: Member loan application and approval. ● Tasks: ○ Implement login and registration features for members. ○ Develop loan application forms and validation processes. ○ Create views for loan approvals and balances. ○ Test functionalities to ensure smooth operations.	Nuraisyah Binti Mohd Zikre
Sprint #3: Staff Module Development ● User Stories Covered: ○ US002: Staff viewing and validating membership applications. ○ US002: Staff validating loan applications. ● Tasks: ○ Create a staff dashboard for membership and loan validation. ○ Develop forms for document validation. ○ Provide download features for lists of members and loans. ○ Conduct unit testing for staff features.	Harini A/P Sangaran
Sprint #4: ALK Module Development ● User Stories Covered: ○ US003: ALK verifying membership applications.	Damiya Aina binti Basir Abd Shammad

○ US003: ALK validating loan applications. ● Tasks: ○ Build an ALK dashboard for reviewing and approving applications. ○ Enable document verification workflows. ○ Test ALK functionalities for accuracy.	
Sprint #5: Final Testing and Deployment ● Tasks: ○ Conduct integration testing for all modules. ○ Optimize system performance. ○ Prepare deployment scripts. ○ Provide user training and manuals.	All members

Table 2:

2.4 User Story Details

This section is a detailed explanation of the user stories described in Persona's user stories. User Story ID for member is US001 whereas the User Story ID for staff is US002. User Story ID for ALK is US003 whereas the User Story ID for staff is US004. US001 is the first User Story, followed by US002, US003 and US004 respectively.

2.4.1 US001: User Story <User Story 1>

Table 3: User Story Description for <Name of Use Case>

User story: Member
ID: US001
User Story Description As a member of Koperasi KADA I want to log in into the system So that I can apply for a loan, view my loan application approval, view loan balance, view savings and financial statements.
Flow of events: 1. The member log in into the system. 2. If the member want to apply for a loan, 2.1 The member needs to click on the loan application button. 2.2 The system will display form by the system for member to fill in the required details 2.3 After filled in all the required details member need to submit the form by agreeing to all the term 3. If the member want to view loan application approval 3.1 The member needs to click on the loan application approval button 3.2 The system will display the status of approval of the application whether it is approved or not 4. If the member want to loan balance 4.1 The member needs to click on the loan balance button. 4.2 The system will display the details of the balance of the loan that needs to be paid by the member. 5. If the member want to view the saving 5.1 The member needs to click on the savings button 5.2 The system will displayed the details of saving from the deduction of the monthly membership fee 6. If the member want to view financial statement 6.1 The member needs to click on the financial statement button. 6.2 The system will displayed the details of the transaction that occur every month based on the salary deduction
Alternative flow 1: If the member does not apply for a loan when view loan application approval • The system will show that the member has not apply for a loan yet
Alternative flow 2: If the member does not apply for a loan when view loan balance • The system will show that the member has not apply for a loan yet
Acceptance Criteria Postcondition: The user is a member of Koperasi KADA Precondition: The loan application applied by member will be recorded in the system Other conditions: The details filled by member will be validated during the process for approval
Exception flow: (if any in the event of error)

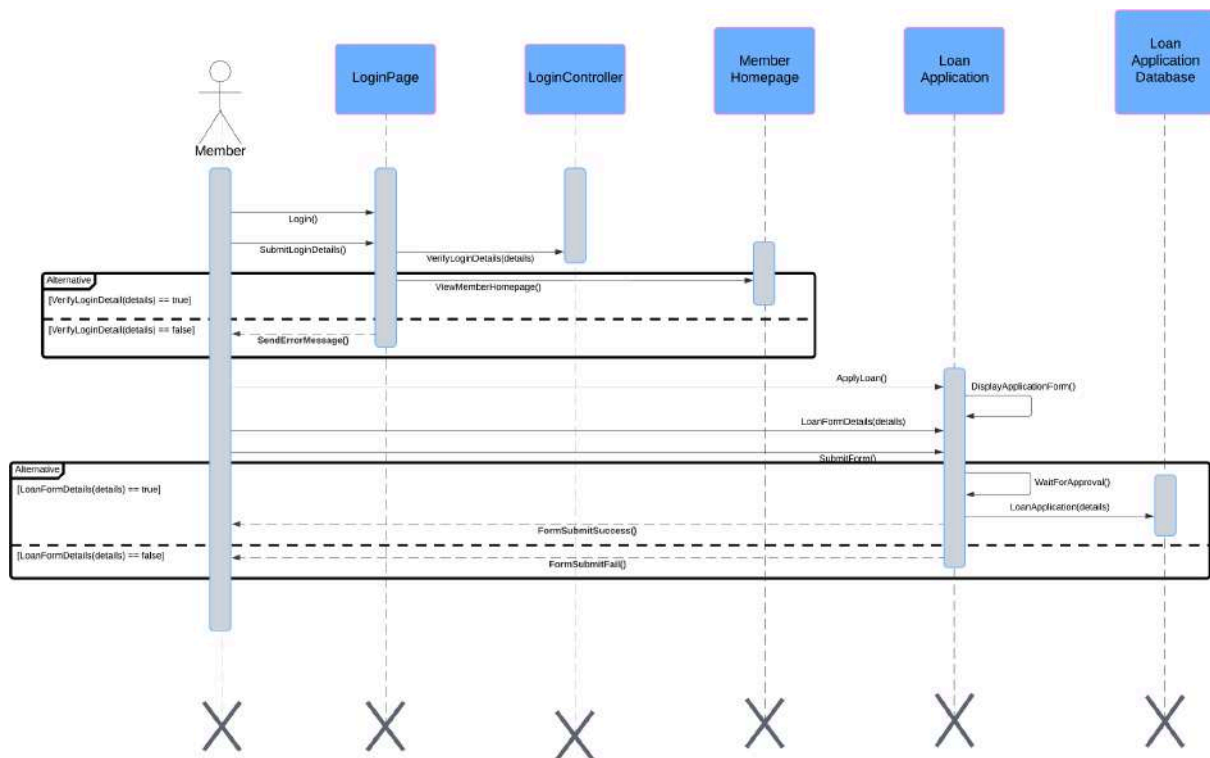


Figure 5: Sequence Diagram for <Name of User Story/Scenario if more than one scenario>

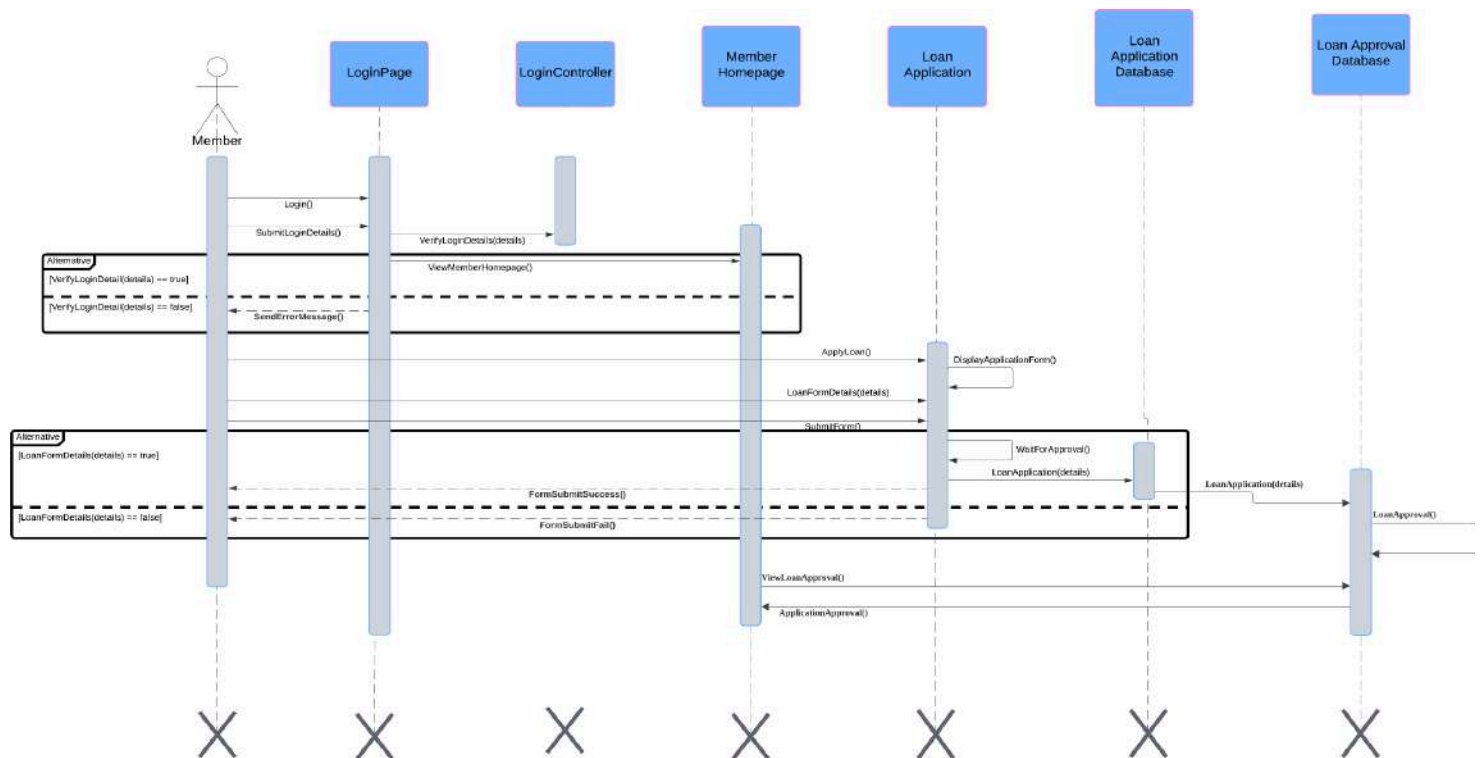


Figure 5: Sequence Diagram for <Name of User Story/Scenario if more than one scenario>

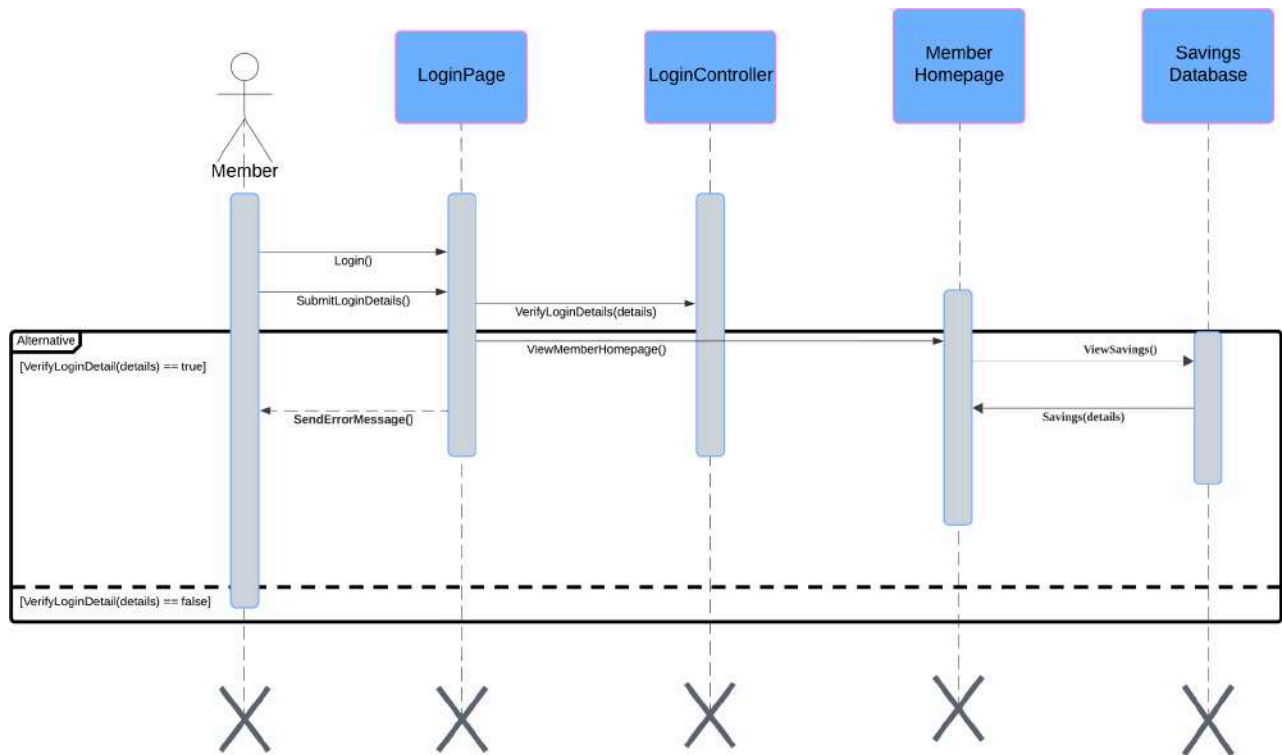


Figure 5: Sequence Diagram for <Name of User Story/Scenario if more than one scenario>

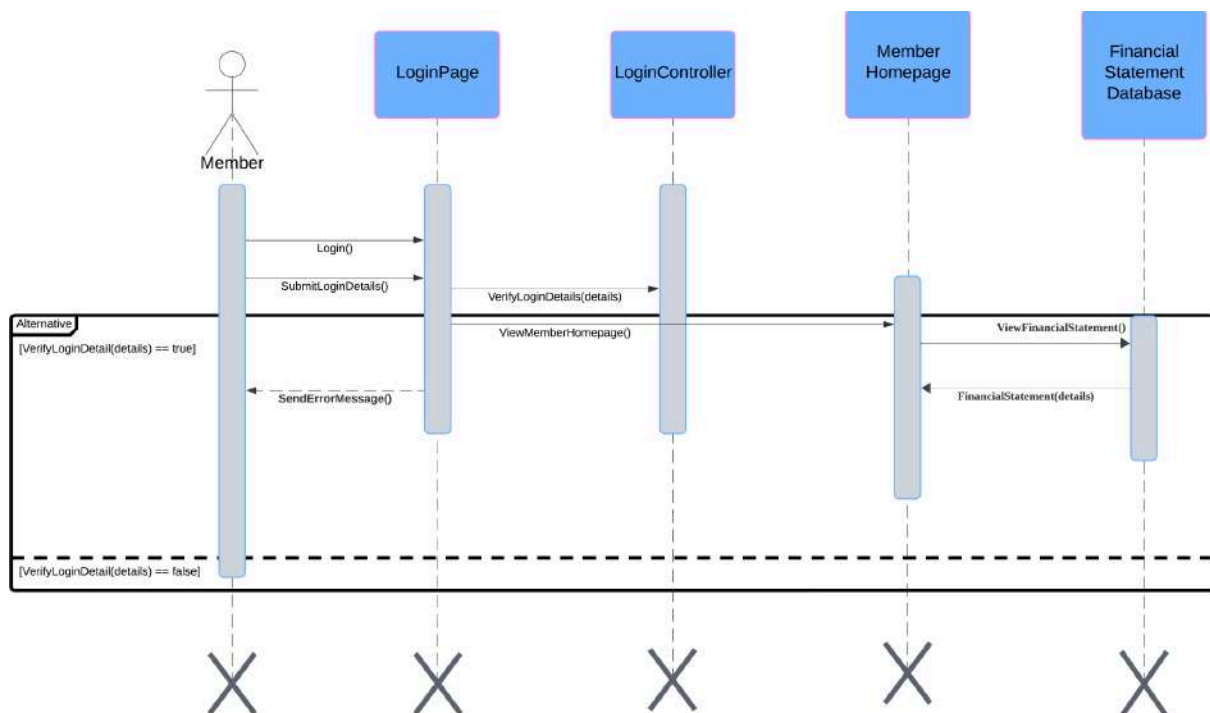


Figure 5: Sequence Diagram for <Name of User Story/Scenario if more than one scenario>

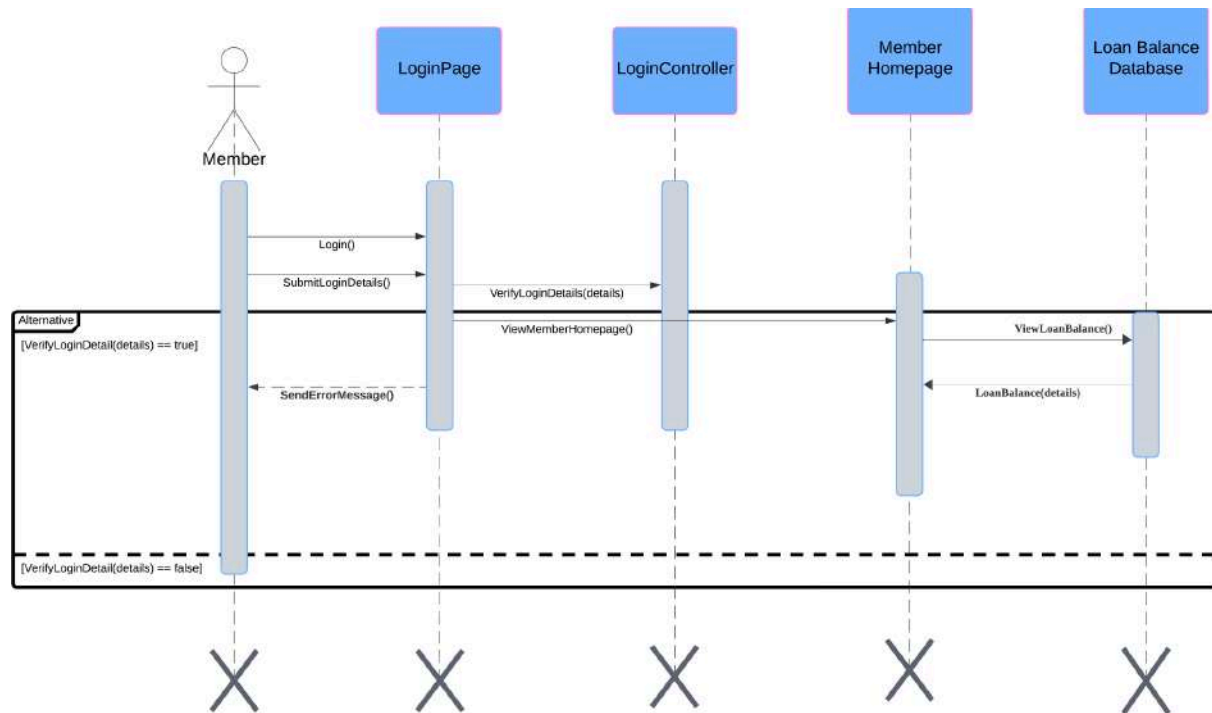


Figure 5: Sequence Diagram for <Name of User Story/Scenario if more than one scenario>

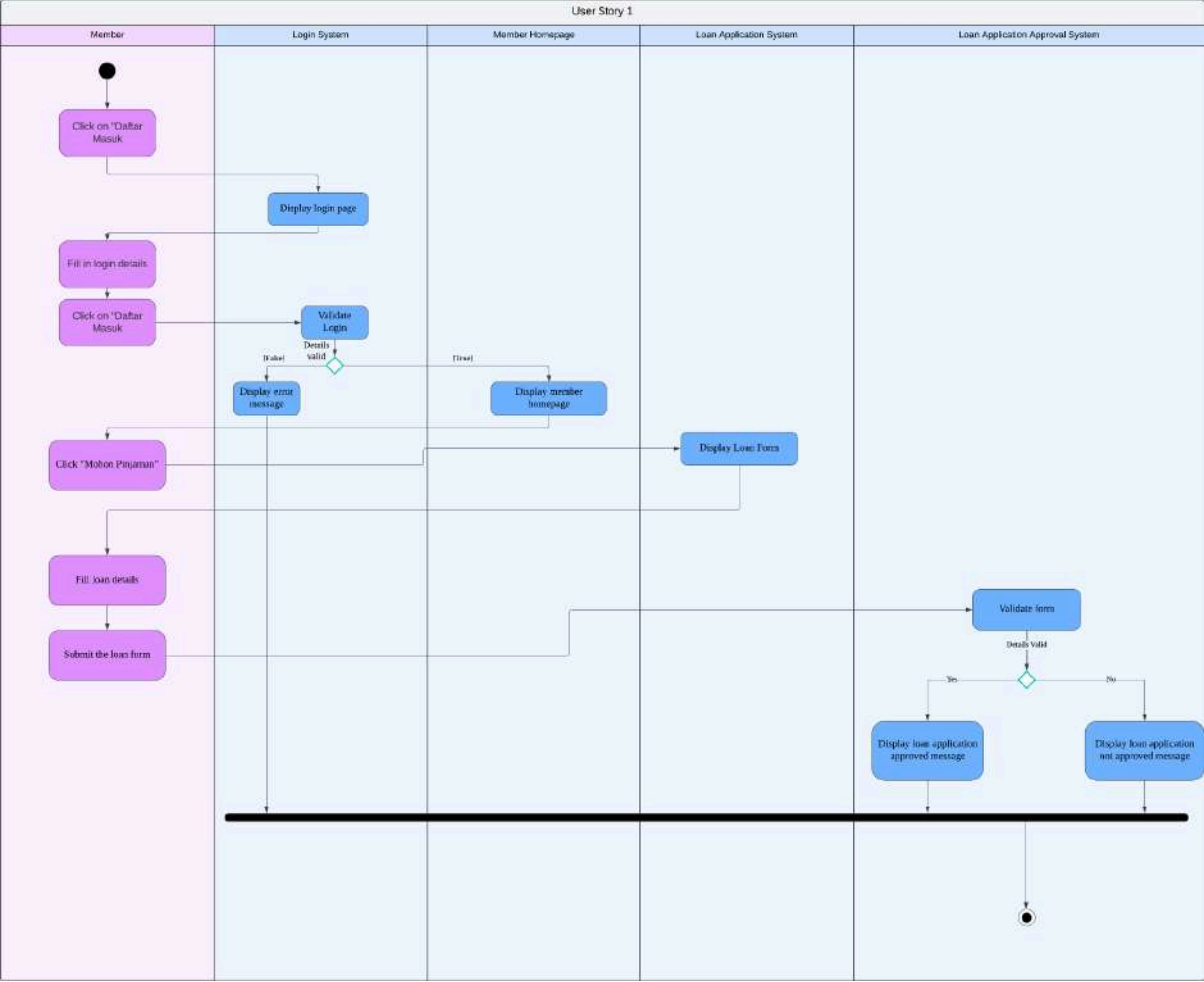


Figure 6: Activity Diagram for <<Name of User Story>>

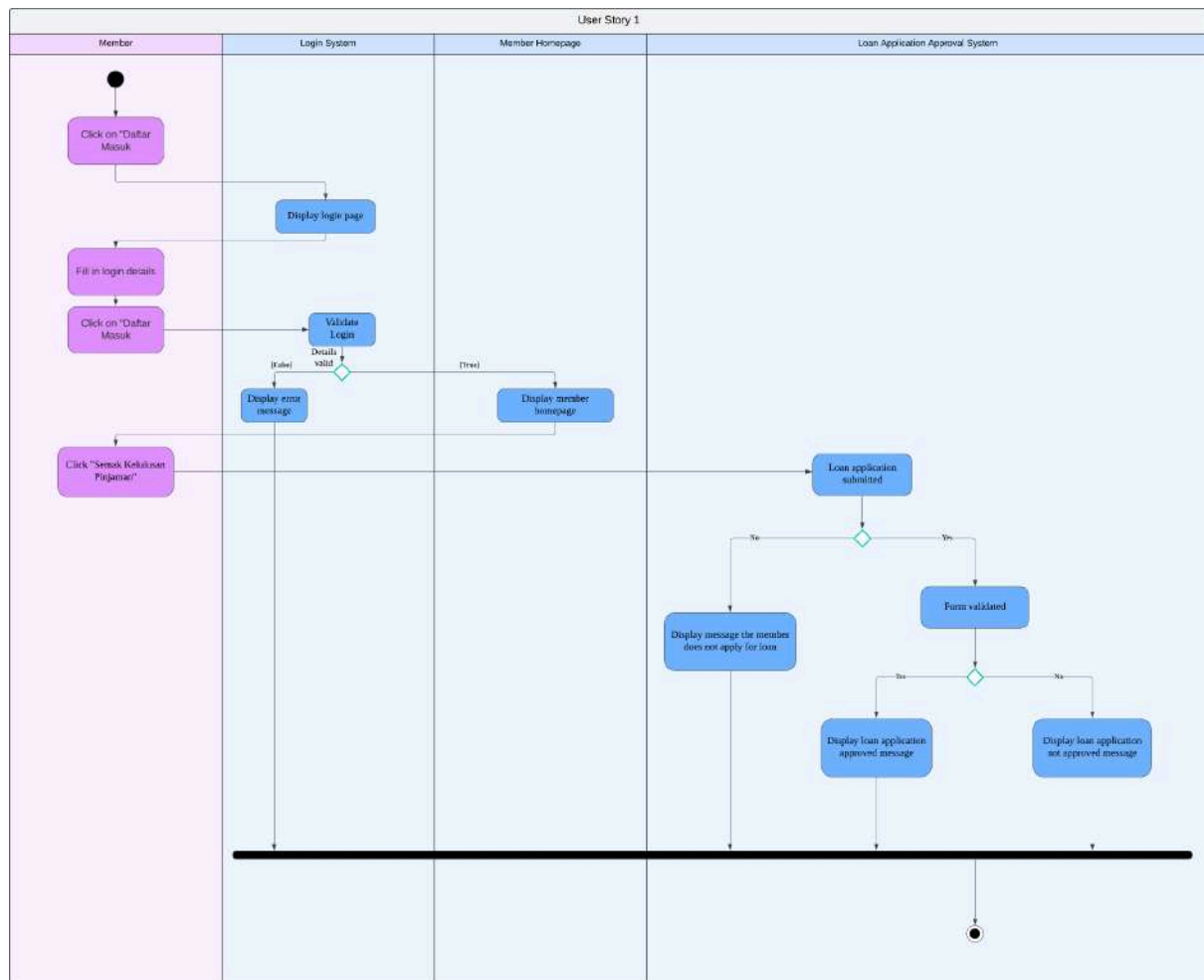


Figure 7: Activity Diagram for <<Name of User Story>>

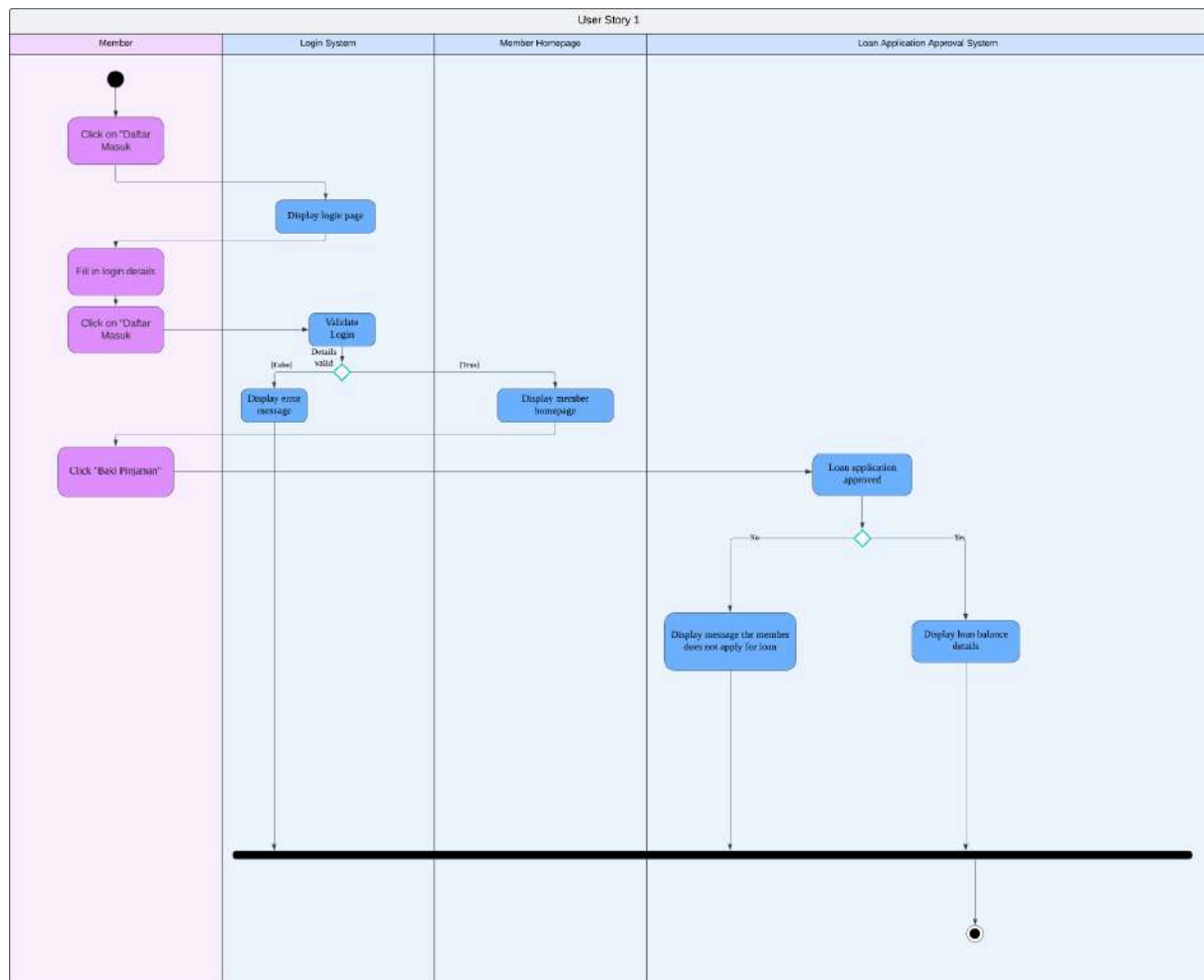


Figure 8: Activity Diagram for <<Name of User Story>>

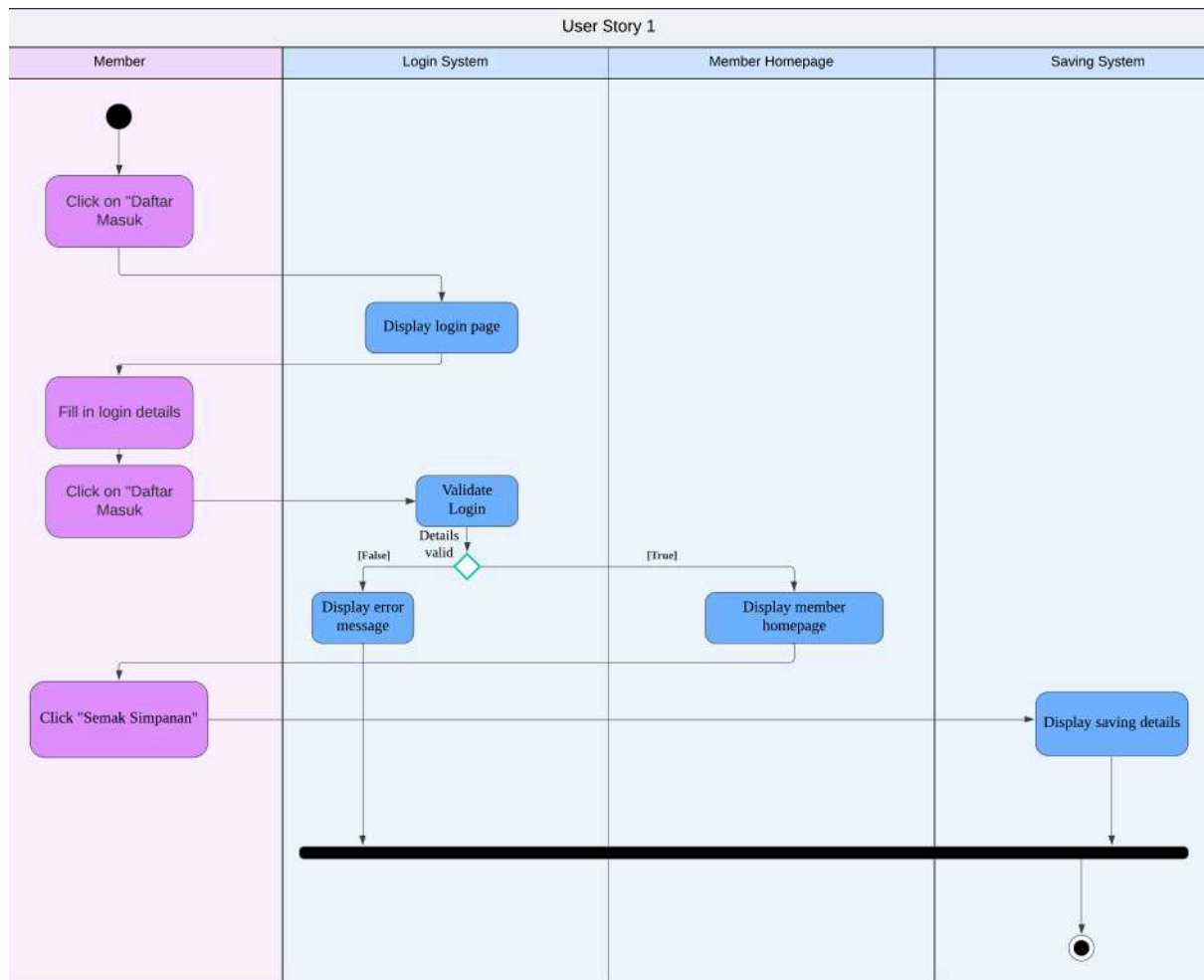


Figure 9: Activity Diagram for <<Name of User Story>>

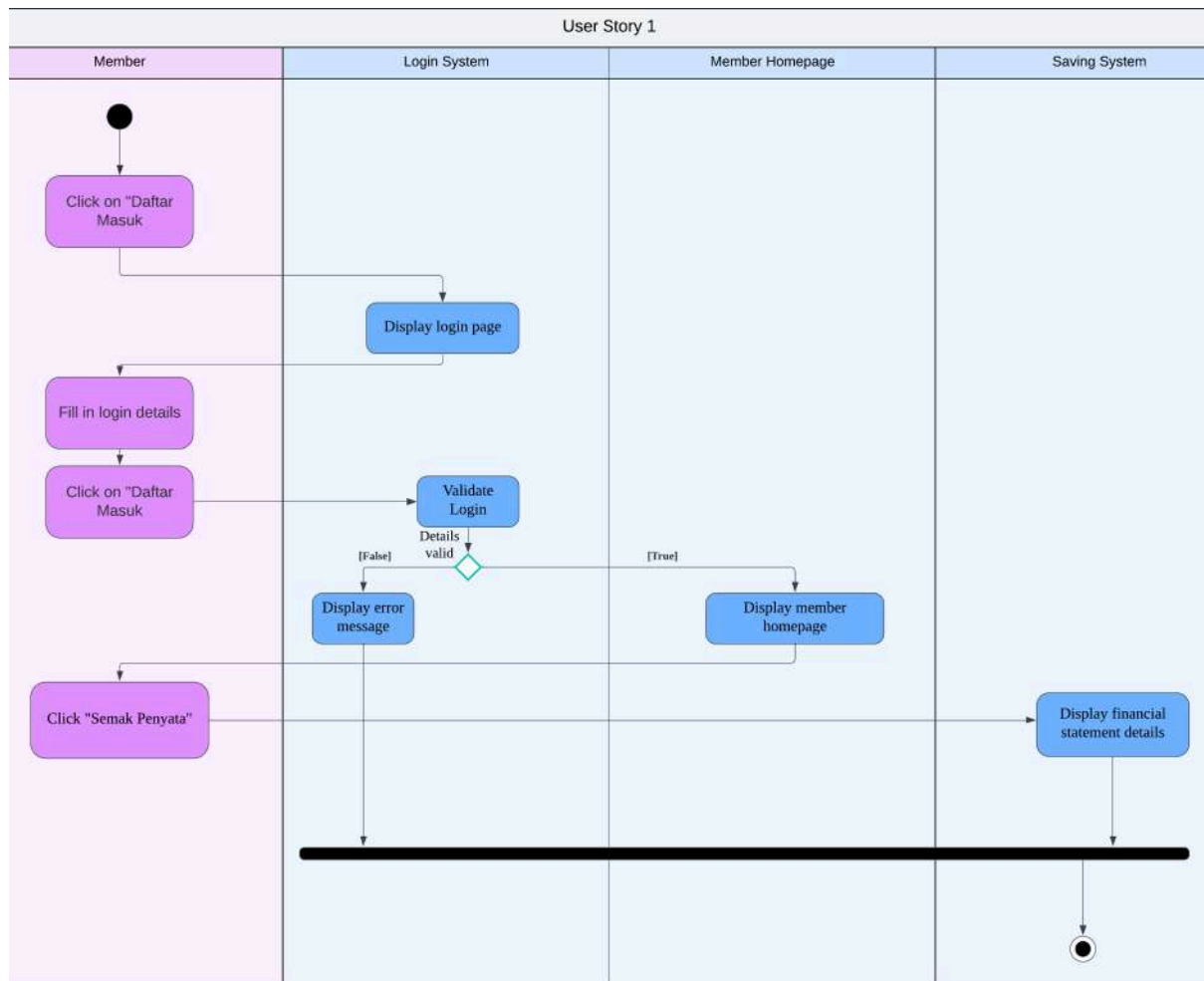


Figure 10: Activity Diagram for <<Name of User Story>>

2.4.2 US002: User Story 2

Table 4: User Story Description for Staff of Koperasi KADA

User story: Staff
ID: US002
User Story Description As a staff of Koperasi KADA I want to log in into the system So that I can view the list of members and list of members with loan and validate the verification documents of non-members who applied for membership and members who applied for a loan.
Flow of events: - The staff logs in into the system. - If the staff wants to view the list of members, - The staff needs to click on the view members list button. - The system will display a list with members' names and ID numbers - If the staff wants to download the members list - The staff have to click on the download button at the bottom of the list

3. If the staff wants to view the list of members with loan
 - 3.1 The staff needs to click on the view members with loan button
 - 3.2 The system will display the members' names, ID numbers and the type of loan they took
 - 3.3 If the staff wants to download the members with loan list
 - 3.3.1 The staff have to click on the download button at the bottom of the list
4. If the staff wants to validate the verification documents for membership application
 - 4.1 The staff needs to click on the membership application list button.
 - 4.2 The system will display the members' names, ID numbers, their verification documents and the status of the validity of the verification documents
 - 4.3 If the staff wants to validify the verification documents
 - 4.3.1 The staff has to click on the verification documents to check the documents
 - 4.3.2 If the documents are as per requirements
 - 4.3.2.1 The staff validates the verification documents
 - 4.3.3 Else
 - 4.3.3.1 The staff does not approve the verification documents
 - 4.3.4 The validity status of the verification documents is updated
 - 4.4 If the staff wants to update the validity of the documents
 - 4.4.1 The staff has to click on the validity status of the verification documents
 - 4.4.2 If the staff wants to validate the verification documents
 - 4.4.2.1 The staff has to click on validate button
 - 4.4.2.2 The staff has to submit the form
 - 4.4.2.3 The validity status of the verification documents is updated
 - 4.4.3 Else
 - 4.4.3.1 The staff has to click on disapprove documents button
 - 4.4.2.2 The staff has to submit the form
 - 4.4.2.3 The validity status of the verification documents is updated
5. If the staff wants to validate the verification documents for loan application
 - 5.1 The staff needs to click on the loan application list button.
 - 5.2 The system will display the members' names, ID numbers, their verification documents and the status of the validity of the verification documents
 - 5.3 If the staff wants to validify the verification documents
 - 5.3.1 The staff has to click on the verification documents to check the documents
 - 5.3.2 If the documents are as per requirements
 - 5.3.2.1 The staff validates the verification documents
 - 5.3.3 Else
 - 5.3.3.1 The staff does not approve the verification documents
 - 5.3.4 The validity status of the verification documents is updated
 - 5.4 If the staff wants to update the validity of the documents
 - 5.4.1 The staff has to click on the validity status of the verification documents
 - 5.4.2 If the staff wants to validate the verification documents
 - 5.4.2.1 The staff has to click on validate button

5.4.2.2 The staff has to submit the form 5.4.2.3 The validity status of the verification documents is updated 5.4.3 Else 5.4.3.1 The staff has to click on disapprove documents button 5.4.2.2 The staff has to submit the form 5.4.2.3 The validity status of the verification documents is updated
Alternative flow 1: If the staff fails to login to the system The system will show that login is unsuccessful
Acceptance Criteria Precondition: The user is a staff of Koperasi KADA

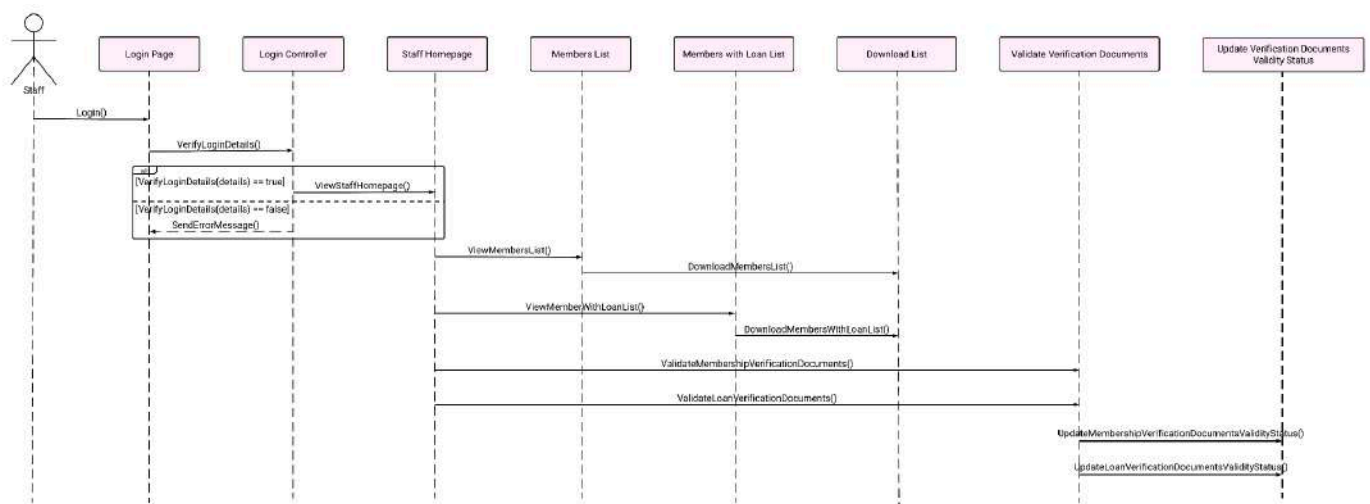


Figure 11: Sequence Diagram for US002 User Story

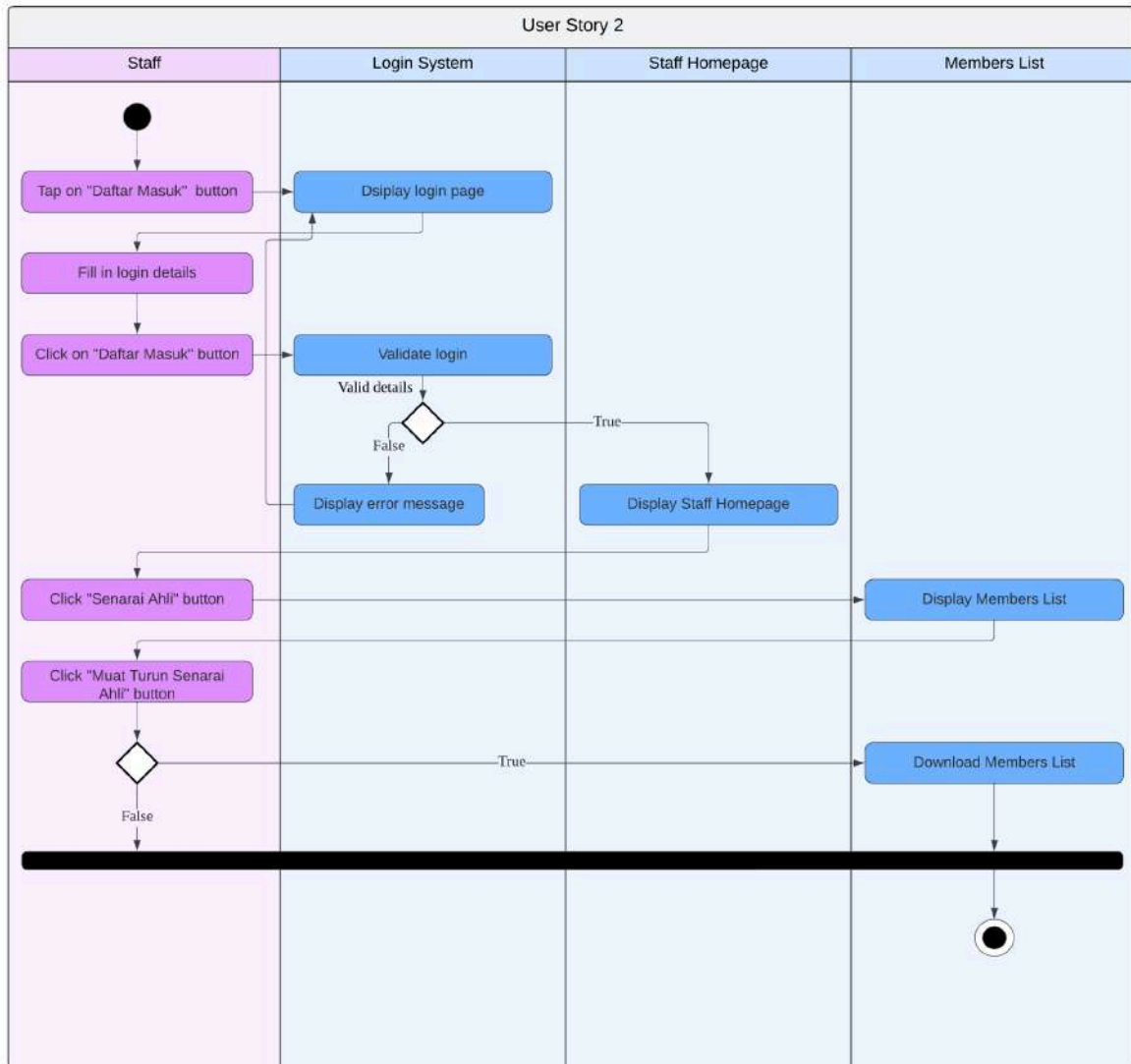


Figure 12: Activity Diagram for when Staff Chooses to View Members List

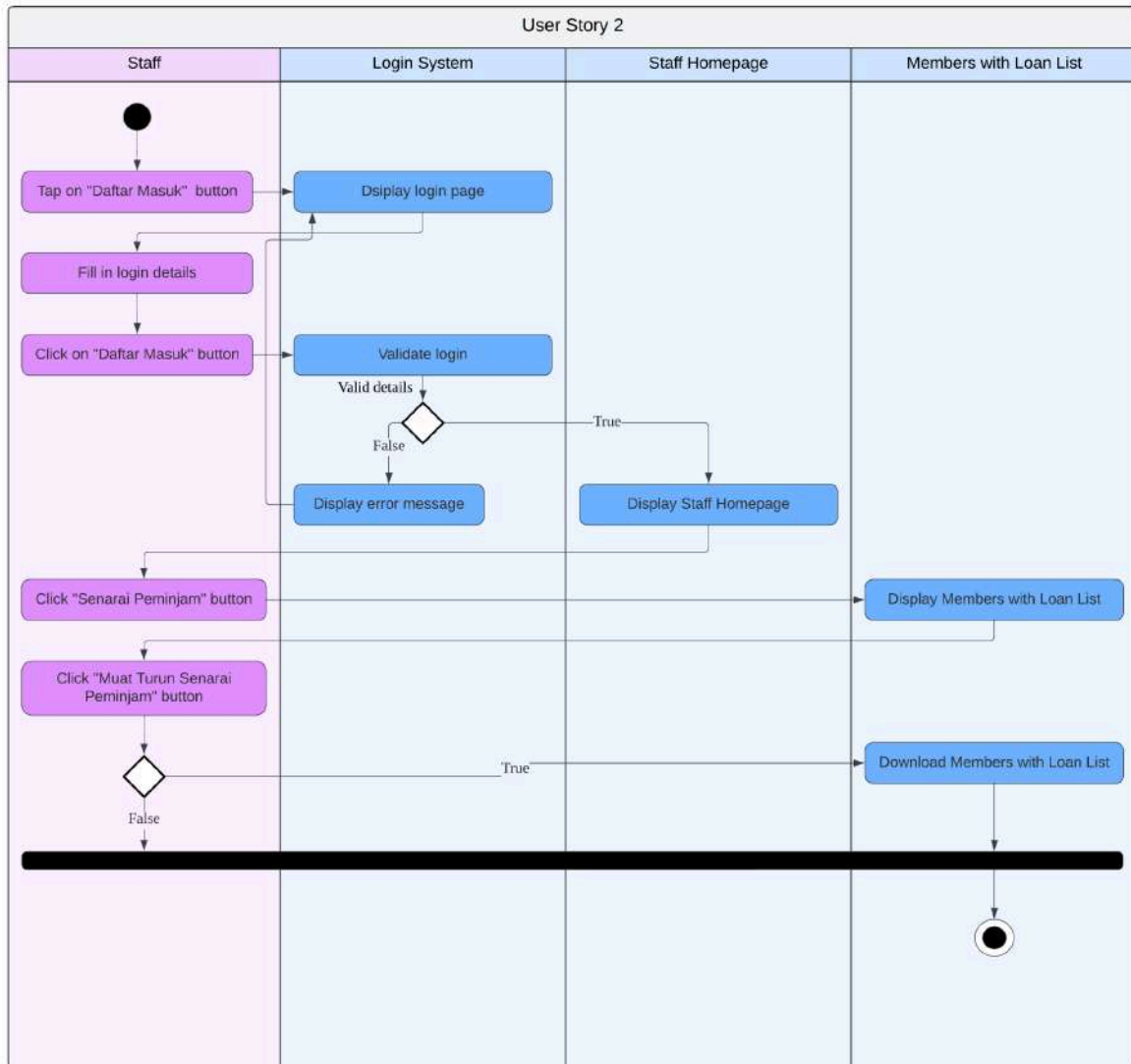


Figure 13: Activity Diagram for when Staff Chooses to View Members with Loan List

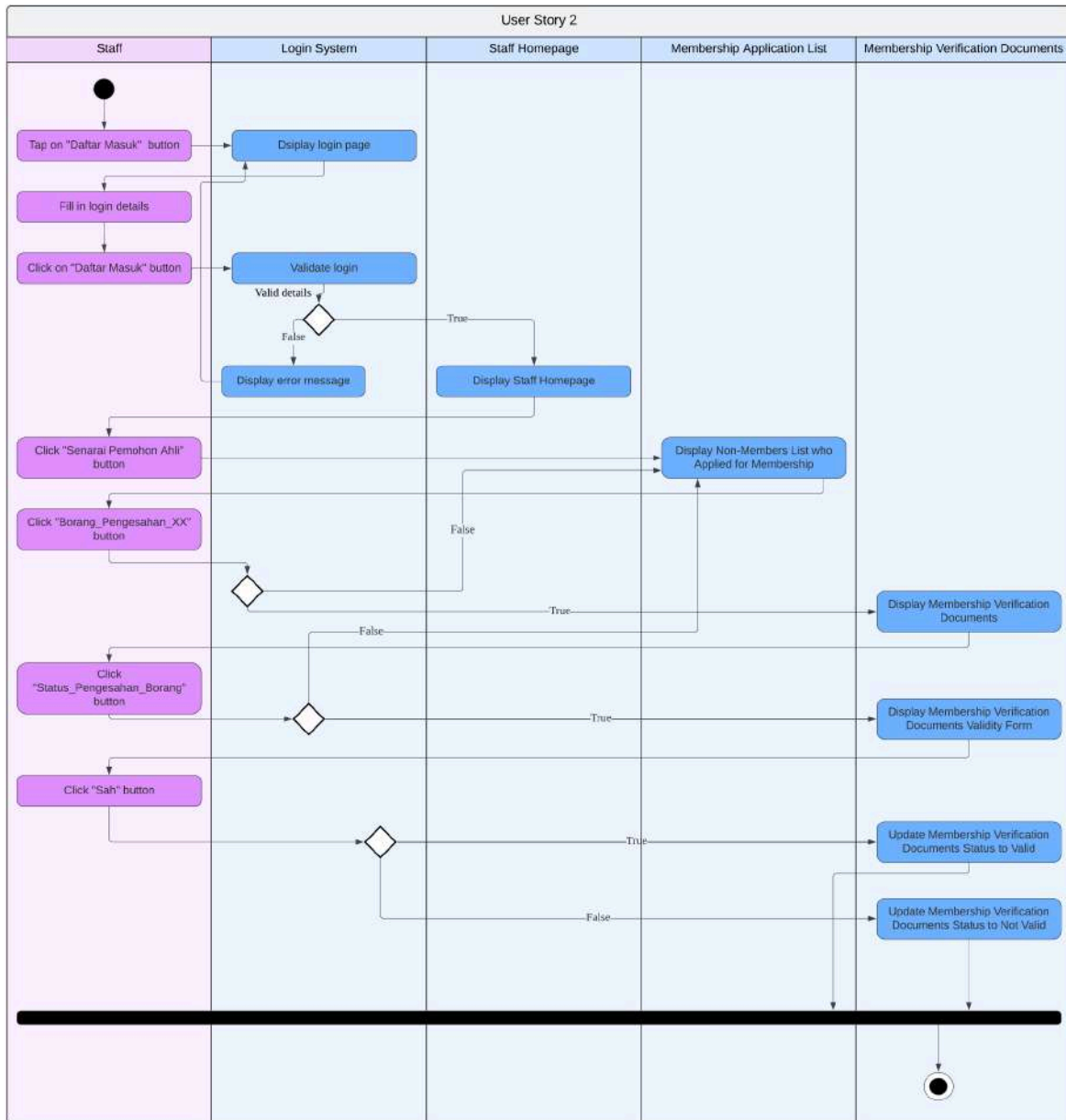


Figure 14: Activity Diagram for when Staff Chooses to View Membership Application List by Non-Members

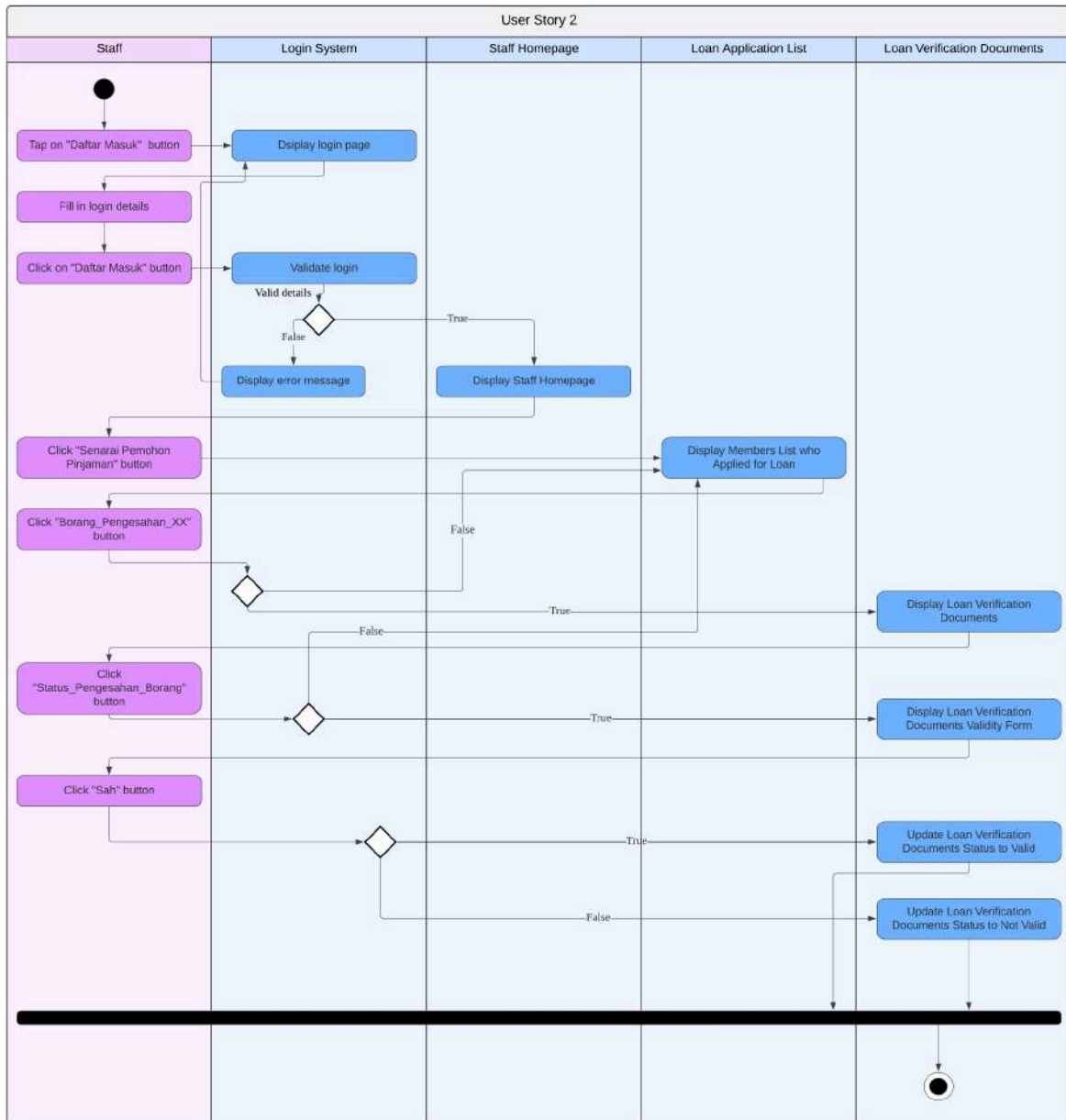


Figure 15: Activity Diagram for when Staff Chooses to View Loan Application List by Members

2.4.3 US003: User Story 3

Table 2.1: User Story Description for US003

User story: ALK module
ID: US003
User Story Description As a high committee member (ALK) of Koperasi KADA, I want to verify membership and loan applications, So that I can ensure the eligibility of applicants and approve valid requests.
Flow of events: 1. The high committee member logs into the Koperasi KADA system by entering their ID and password. 2. The high committee member accesses the homepage with features to view: - Membership applications from non-members. - Loan applications from members. 3. The high committee member clicks on the list of non-members applying for membership to view their application details. 4. The high committee member reviews the verification documents and validates the membership application. 5. The high committee member clicks on the list of members applying for loans to view their application details. 6. The high committee member reviews the loan type, verification documents, and other details to validate the loan application. 7. The system updates the status of the application (approved/rejected). 8. The high committee member may update the status later if needed.
Alternative flow: 1. Invalid Application Documents: The system prompts the high committee member to mark the application as rejected and notifies the applicant of the rejection. 2. Incomplete Review: The high committee member exits the application review and marks it as pending. The system saves the review progress.
Acceptance Criteria Postcondition: The status of the membership or loan application is updated (approved/rejected/pending). Precondition: The high committee member is logged into the system and has access to application details.

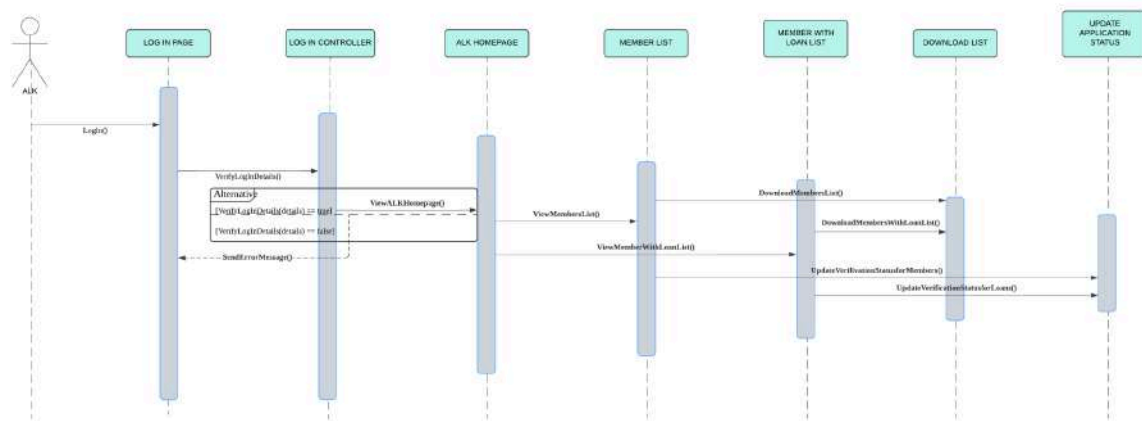


Figure 2.5: Sequence Diagram for US003 User Story

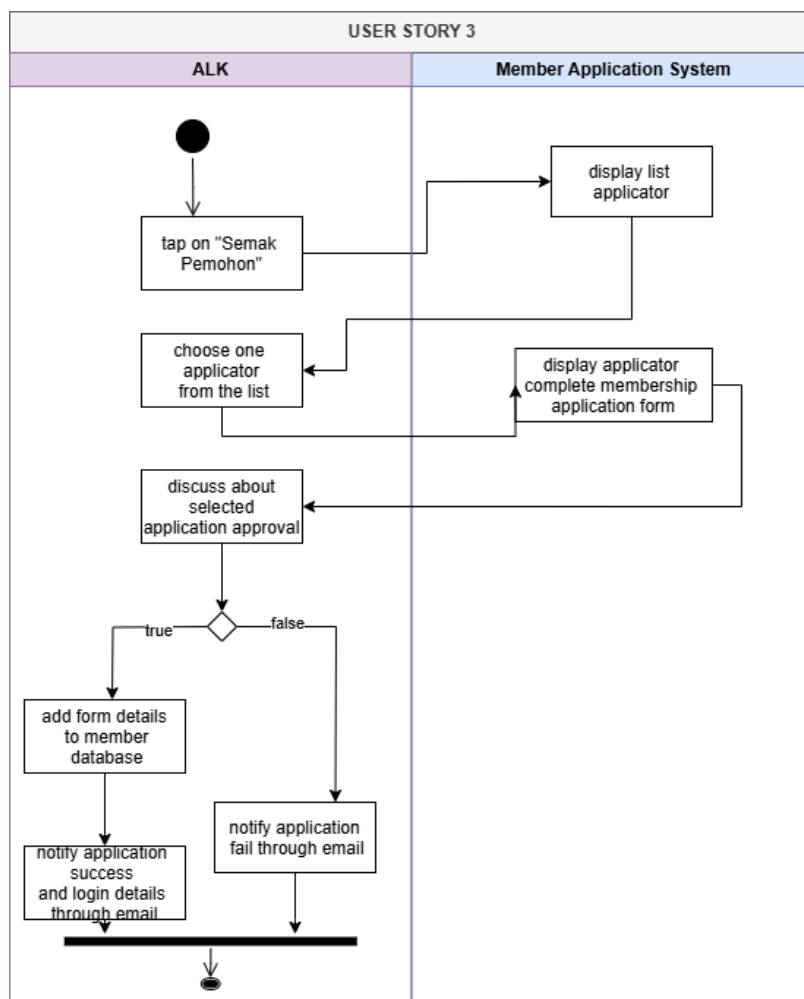


Figure 2.5: Activity Diagram for ALK approved member application

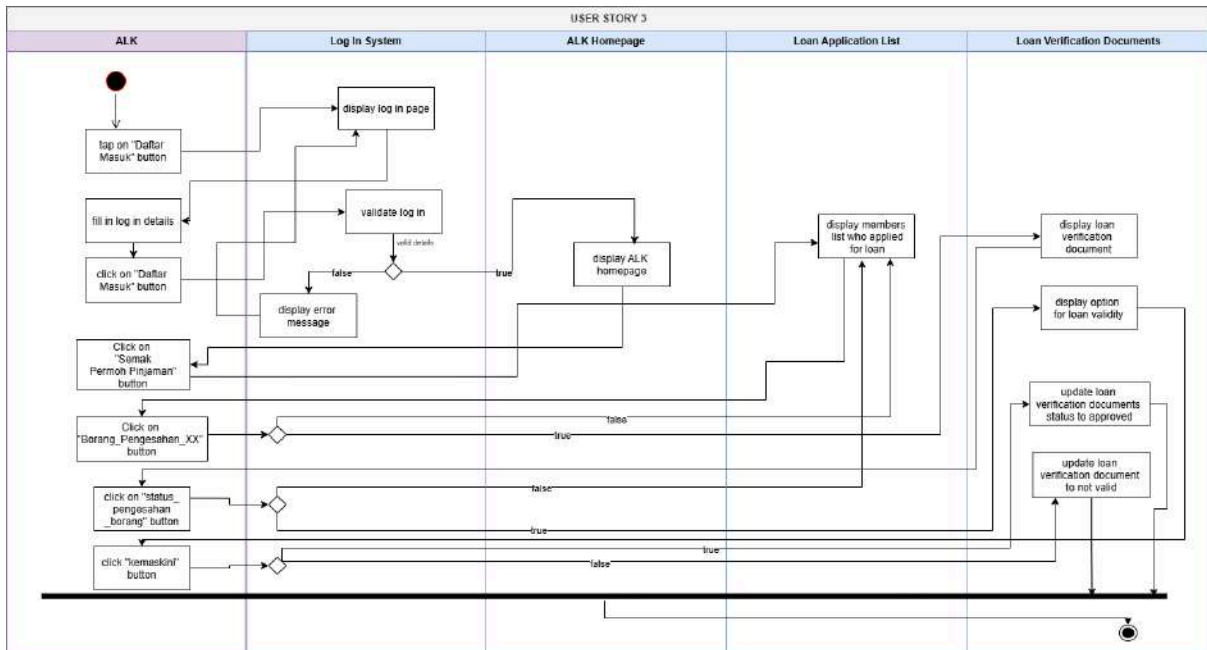


Figure 2.5: Activity Diagram for ALK approved loan application

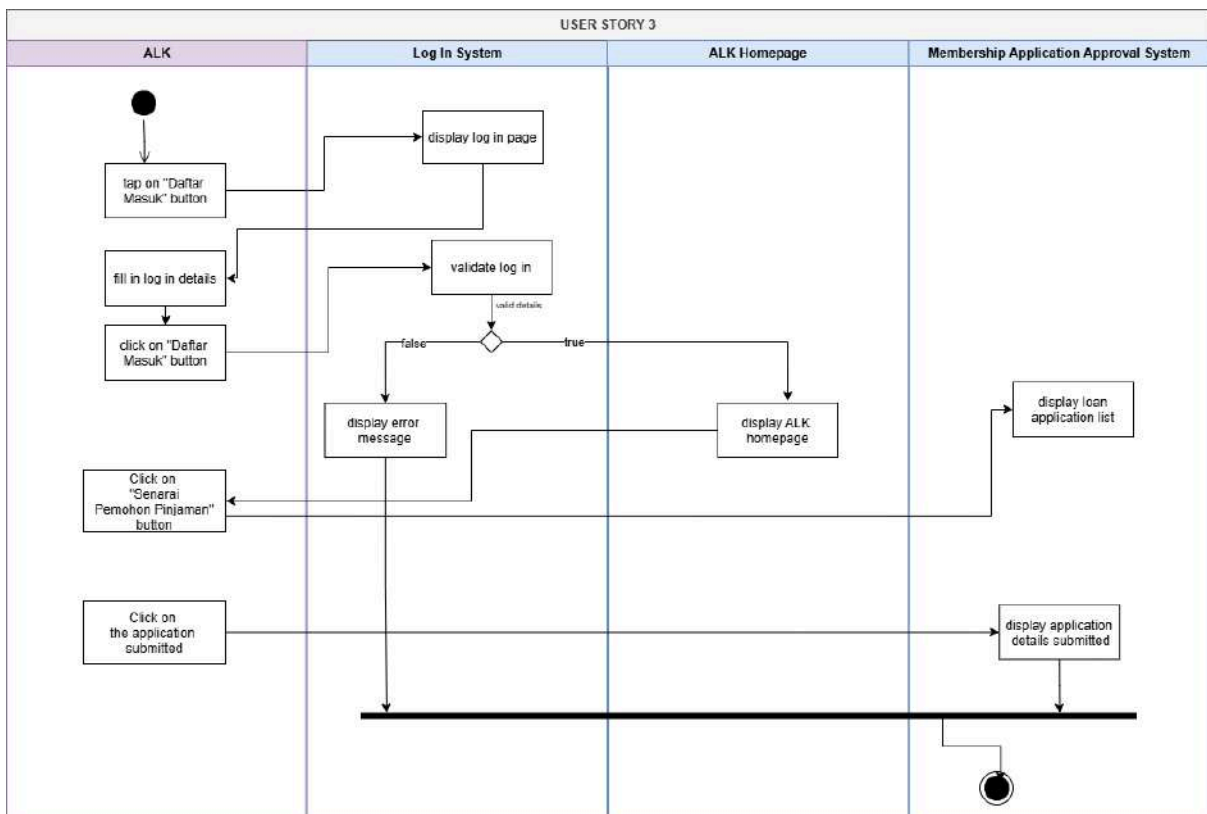


Figure 2.5: Activity Diagram for ALK view validated loan application

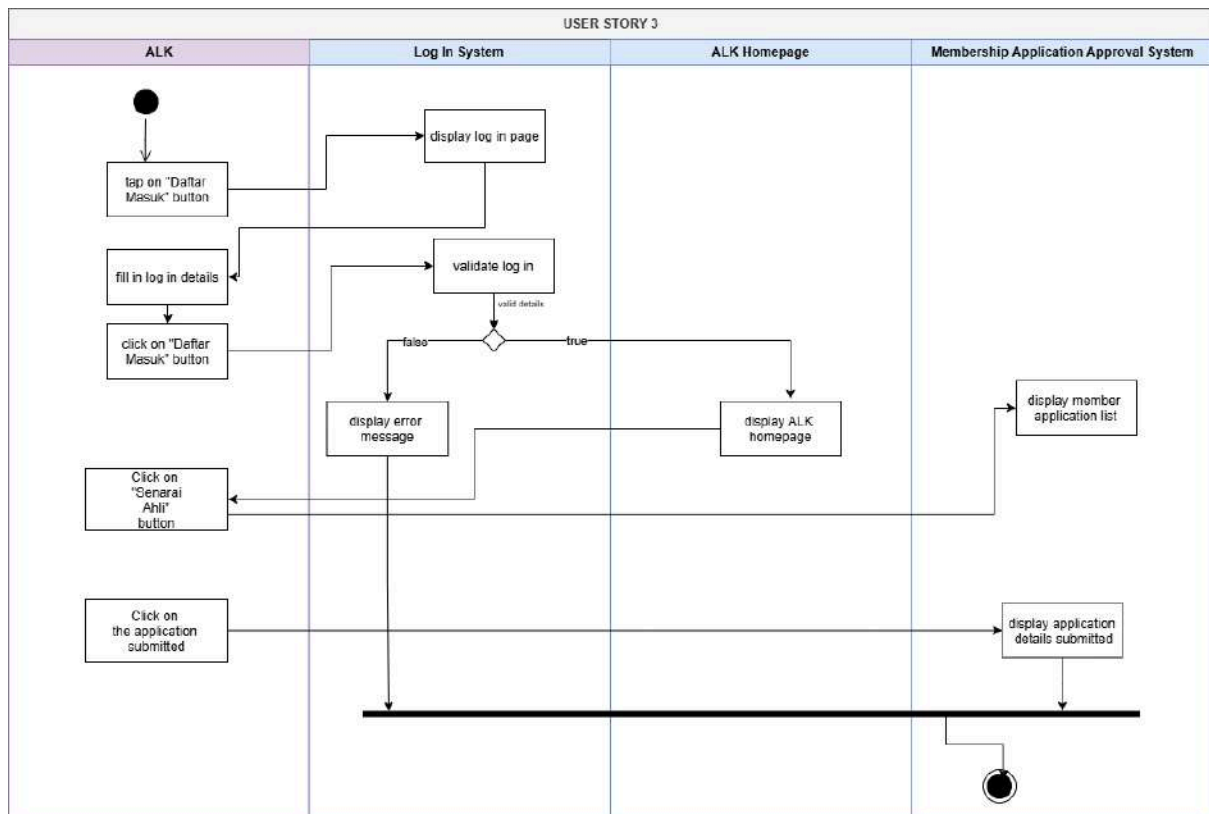


Figure 2.5: Activity Diagram for ALK view validated member application

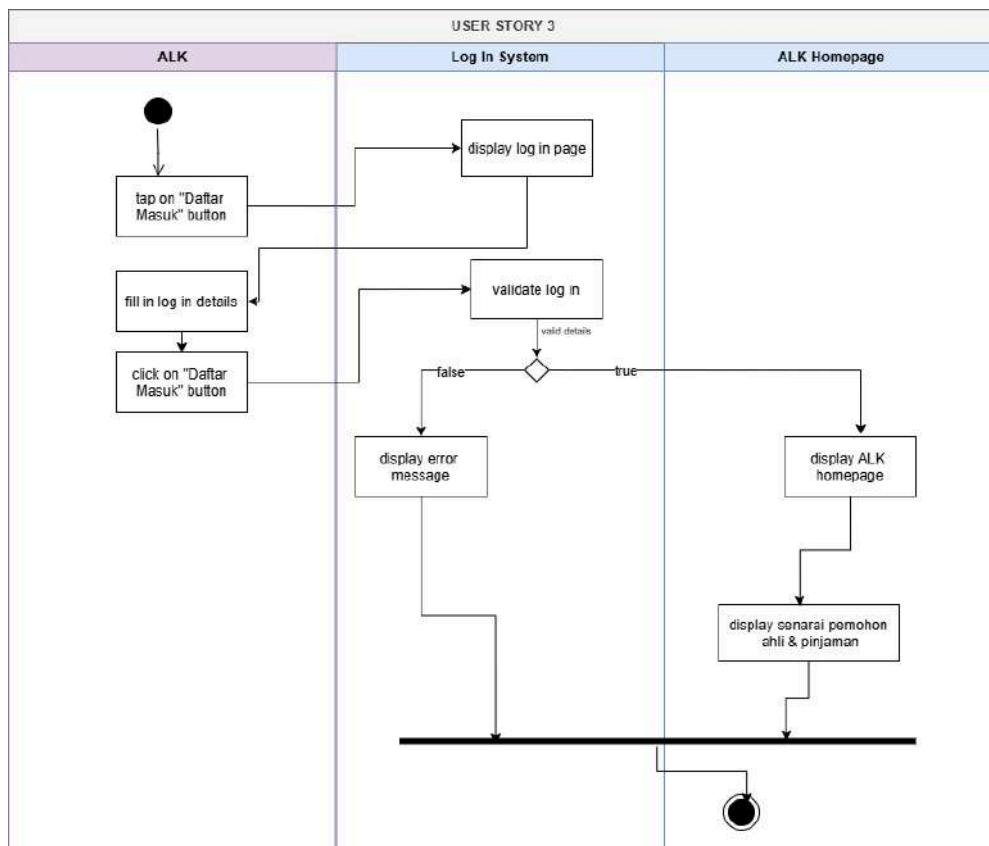


Figure 2.5: Activity Diagram for view ALK menu

2.4.4 US004 User Story 4

Table 2.1: User Story Description for US004

User story: Non-member module
ID: US004
User Story Description As a non-member of Koperasi KADA I want to apply for membership So that I can login into the system
Flow of events: 1. The non-member click on the apply membership to access the membership application form page 2. The non-member fill in all the details needed in the form 3. The non-member submit the form and wait for approval 4. The non-member get an email with login details when application approved 5. The non-member accessed login page 6. The non-member input their login details given from email 7. The system verified login details and show the system homepage
Alternative flow: Application failed: Email sent to non-user notify them and tell them to reapply if they want. Invalid login details: The system prompts the user to re-enter login details or contact the staff for further assistance.
Acceptance Criteria Postcondition: The non-member is now a member of Koperasi KADA. Precondition: The non-member is inside the Koperasi KADA homepage.

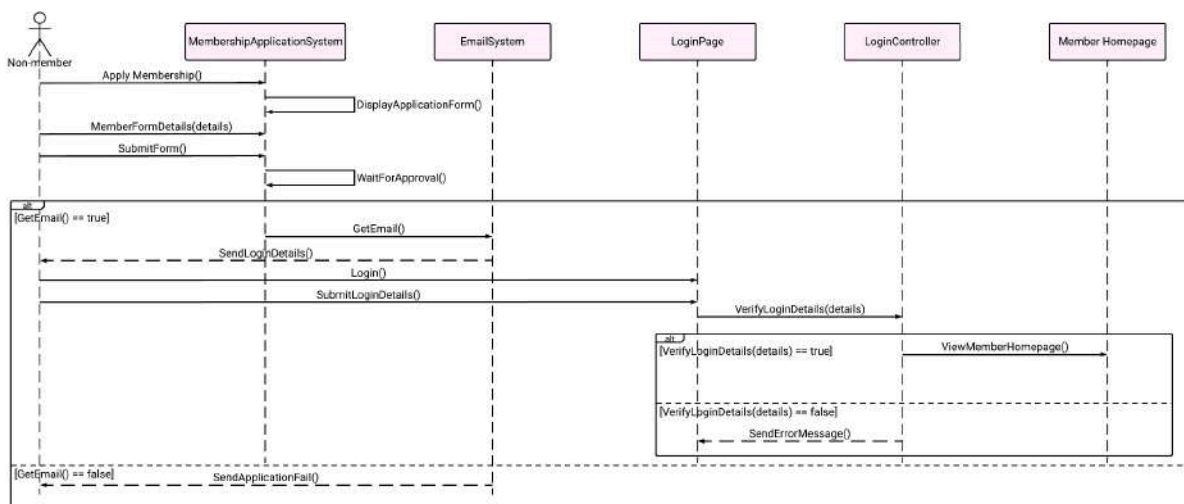


Figure 2.5: Sequence Diagram for US004 User Story

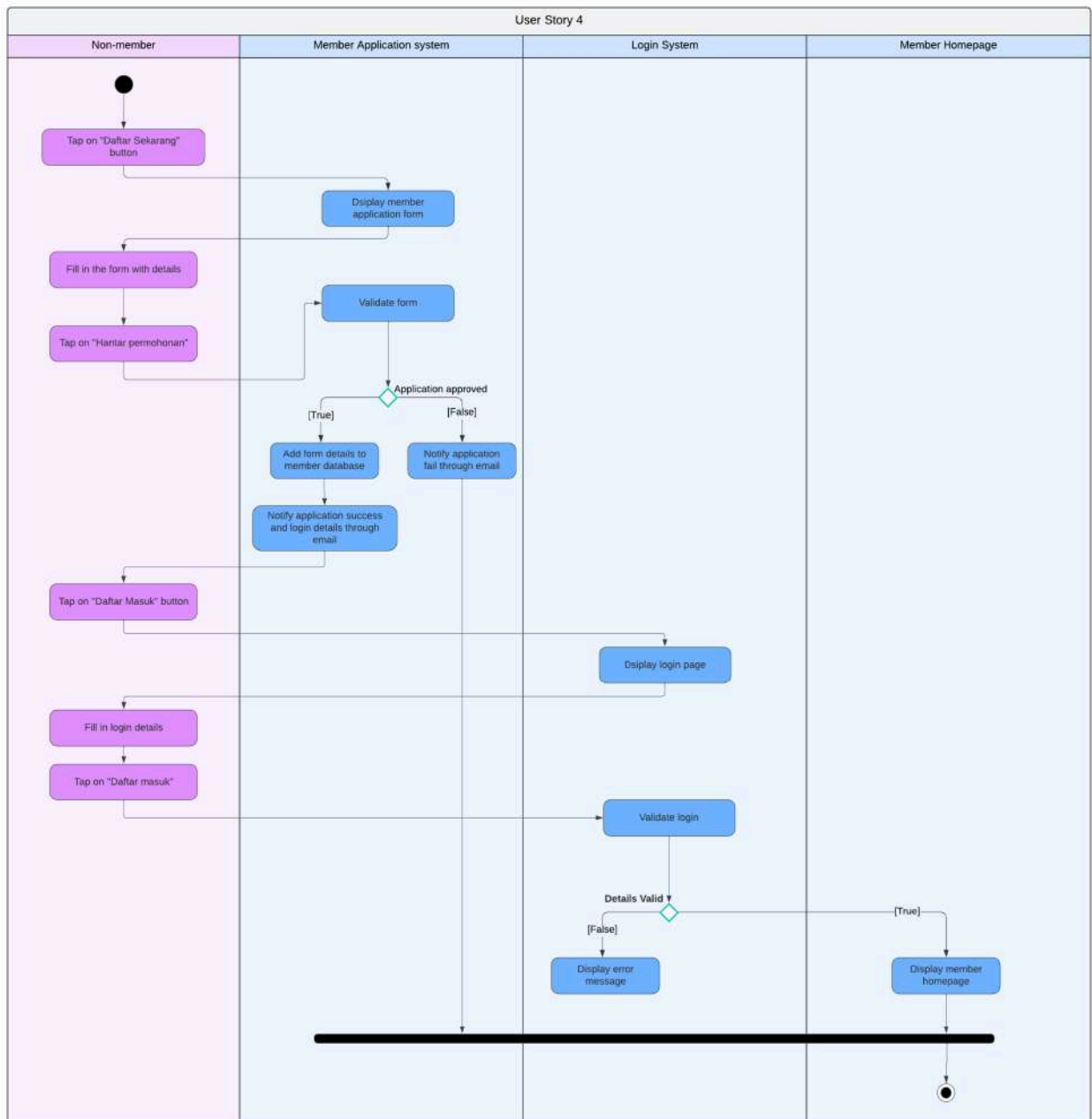


Figure 2.2: Activity Diagram for US004 User Story

2.5 Performance and Other Requirements

In this section, we will discuss the performance requirements and software system attributes of the proposed system for Koperasi KADA. Performance requirements include response time, throughput, capacity and availability whereas software system attributes involve usability, reliability, maintainability, portability, compatibility, security, safety, legal and regulatory, and lastly environmental impact.

Performance Requirements

1. **Response Time** : The system is designed to respond within 3 seconds of user input, ensuring a fast and efficient experience. The response time minimizes delays and ensures smooth navigation throughout the system.
2. **Throughput** : The system is designed to handle up to 30 requests simultaneously. This throughput ensures the proposed system can efficiently support peak usage periods or multiple simultaneous operations without experiencing slowdowns.
3. **Capacity** : The system is capable of managing up to 30 active users at the same time while handling a data load of approximately 10,000 entries. This capacity ensures the system can accommodate typical user activity without compromising system performance.
4. **Availability** : The system is operational 24 hours a day, 7 days a week. Thus, users can be worry free of having a time constraint. Users can complete tasks or access information whenever needed.

Software System Attributes:

1. **Usability**: The system must have an intuitive user interface, enabling users to navigate the platform with minimal training.
2. **Reliability**: The system must perform consistently without errors or crashes, ensuring stable operation.
3. **Maintainability**: The codebase should be modular and well-documented to allow easy updates and enhancements.
4. **Portability**: The system must operate on multiple web browsers and mobile devices without compatibility issues.
5. **Compatibility**: It should integrate seamlessly with existing KADA databases and infrastructure.
6. **Security**: The system must employ encryption and secure authentication to protect sensitive data and ensure compliance with regulations.
7. **Safety**: The system must operate safely, preventing any data corruption or harm to the environment.
8. **Legal and Regulatory Compliance**: The system must comply with relevant data protection laws such as Malaysia's PDPA.
9. **Environmental Impact**: The system should utilize optimized resources to minimize server load and energy consumption.

2.6 Design Constraints

In this section, we will point out some design constraints we found regarding the system that is about to be built, KADA Corporation System when being implemented into the organization and explain it in detail.

Compatibility constraints:

The system must seamlessly integrate with the existing workflows and infrastructure, including backend systems and databases, to maintain compatibility. The existing stored data needs to be migrated without loss or corruption.

Online accessibility constraints:

The system should support diverse devices and browser, ensuring accessibility for all users within the organization

Real-time features constraints:

The system has real-time capabilities which is crucial for providing instant updates on loan statuses, membership applications, and savings or loan balances. It should ensure the response time is less than 2 seconds and actions under normal server load conditions

Security constraints:

The system must have robust security measures, such as encryption and user authentication, must be implemented to protect sensitive financial and personal data, while ensuring compliance with data privacy regulations like PDPA (Personal Data Protection Act) in Malaysia. A role-based access control (RBAC) system needs to be implemented to prevent unauthorized access.

Scalability constraints:

The system must be scalable to accommodate a projected 15% growth in user load, reflecting the organization's expected expansion in operations.

Hardware constraints:

The system will require a sufficient storage of cloud-based hosting to handle every user request. The system must be compatible with various devices with varying specifications. A compatible peripheral device is needed for administration in order to expand its capabilities.